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Scalable Kernel Methods for Sequentially Ordered Data

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I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated. I have only used the resources given in the list of references.

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Abstract

Sequentially ordered data naturally arise in a multitude of Data Science applications, thus undermining the need for scalable Machine Learning algorithms. This thesis aims to build a mathematically rigorous bridge between efficient kernel methods for sequentially ordered data and the signature transform, which is a non-commutative power series in the tensor algebra whose coefficients can be seen in close analogy to monomials. We begin with a self-contained review of the signature transform from a purely algebraic perspective and then connect these objects to kernel methods via the signature kernel, thereby generalizing polynomial kernels from static to sequentially ordered data. Additionally, we provide a more practically usable discretized version of the signature kernel.

To overcome the quadratic cost in the number of sequences incurred by Gram matrix computations, we analyse two prominent dimensionality-reduction techniques that yield asymptotically linear training time in the number of sequences. We first combine Random Fourier Features with the discretized signature kernel, which are based on a Monte-Carlo approximation of a spectral measure that arises due to Bochner's Theorem. We provide one theoretical contribution by focusing on tensor random projections; using the Khatri-Rao product of very-sparse random projections we prove pointwise convergence with mixed tails and show fair speed-up compared to the iid Gaussian case presented in [TOS25] for high-dimensional sequential data.

Zusammenfassung

Sequentiell geordnete Daten treten in zahlreichen Anwendungen der Data Science auf und machen damit skalierbare Machine-Learning-Algorithmen erforderlich. Die nachfolgende Bachelorarbeit hat zum Ziel, eine mathematisch rigorose Brücke zwischen effizienten Kernel-Methoden für sequentiell geordnete Daten und der Signature Transform zu bilden. Letztere ist eine nichtkommutative Potenzreihe in der Tensoralgebra, deren Koeffizienten in enger Analogie zu Monomen stehen. Wir beginnen mit einem kurzen theoretischen Einblick in die Funktionsweise der Signature Transform aus rein algebraischer Sicht und verknüpfen diese Objekte anschließend über den Signature-Kernel mit Kernel-Methoden; damit werden polynomielle Kernel-Methoden von statischen auf sequenziell geordnete Daten verallgemeinert. Zusätzlich gehe wir auf eine in der Praxis nutzbare diskretisierte Variante ein.

Um die quadratischen Laufzeit in der Anzahl der Datenpunkte, die bei Gram-Matrix Berechnungen anfallen, zu vermeiden, analysieren wir zwei prominente Dimensionsreduktionsverfahren, die asymptotisch lineare Trainingszeit in der Anzahl der Datenpunkte ermöglichen. Zunächst kombinieren wir Random Fourier Features mit dem diskretisierten Signature-Kernel; diese basieren auf einer Monte-Carlo-Approximation eines Spektralmaßes, das aus dem Satz von Bochner hervorgeht. Unser theoretischer Beitrag fokussiert sich hingegen auf tensorwertige Zufallsprojektionen: Unter Verwendung des Khatri-Rao Produkts von sehr dünnbesetzten Zufallsprojektionen beweisen wir punktweise Konvergenz mit gemischten Tails und zeigen einen deutlichen Laufzeitvorteil gegenüber den verwendeten unabhängig, normalverteilten Zufallsprojektion aus [TOS25] für hochdimensionale sequentielle Daten.

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Chapter 1
Signature Transform

Sequentially ordered data, naturally arising as time series, appears in various domains, from finance and econometrics to pattern recognition and even astronomy. A stream of sequential information can be interpreted as a function or path in a state-space. Originating in Chen's theory of iterated integrals, the Signature Transform of such a path is a feature-rich, coordinate-free and reparametrization-invariant tensor-based representation with possibly infinitely many elements.

1.1 Preliminaries

In the following we use the notation $[n] := \{1, 2, \ldots, n\}$ for any integer $n \in \mathbb{N}$ with the convention $[0] := \varnothing$.

The algebraic backbone of the Signature Transform are subsets of tensor algebras with group-structure, which will be briefly introduced in the following section.

Definition 1.1 (Tensor Product) Let U, V be two vector spaces. Their *tensor product* is a vector space $U \otimes V$ with a bilinear map $\varphi : U \times V \to U \otimes V$ such that

a) $\mathrm{im}(\varphi) = U \otimes V$,

b) for a bilinear map $\psi : U \times V \to W$ into an arbitrary vector space W there exists a bilinear map $\widehat{\psi} : U \otimes V \to W$ such that $\psi = \widehat{\psi} \circ \varphi$.

The conditions 1.1a) and 1.1b) are in unity referred to as *universal property* of the tensor product [Gre78]. It can be shown that 1.1a) and 1.1b) are equivalent to the single property

b') for a bilinear map $\psi : U \times V \to W$ into an arbitrary vector space W there exists a *unique* bilinear map $\widehat{\psi} : U \otimes V \to W$ such that $\psi = \widehat{\psi} \circ \varphi$.

The tensor product is well-defined and can henceforth be written as $u \otimes v = \varphi(u, v)$ for any $u \in U, v \in V$. It can also be extended inductively to more than two vector spaces.

The universal property characterizes the tensor product in the sense that it is the up to an isomorphism unique vector space having this property, which follows immediately for two tensor products by applying the universal property of the first tensor product to the second and vice versa to check that their concatenation is the identity.

Proposition 1.2 *Let U, V be finite-dimensional vector spaces with $\dim(U) = n$, $\dim(V) = m$. Further, define a basis $\{u_i\}_{i \in [n]}$ of U and take a basis $\{v_j\}_{j \in [m]}$ of V. Then $\{u_i \otimes v_j\}_{i \in [n], j \in [m]}$ is a basis of $V \otimes W$.*

Proof. For arbitrary coefficients $\alpha_i, \beta_j \in \mathbb{R}$ with $i \in [n]$, $j \in [m]$, define $u = \sum_{i=1}^{n} \alpha_i u_i,$, $v = \sum_{j=1}^{m} \beta_j v_j$, then by the bilinearity of the tensor product

$$u \otimes v = \left(\sum_{i=1}^{n} \alpha_i u_i \right) \otimes \left(\sum_{j=1}^{m} \beta_j v_j \right) = \sum_{i=1}^{n} \sum_{j=1}^{m} \alpha_i \beta_j (u_i \otimes v_j).$$

Thus, $\operatorname{span}(\{u_i \otimes v_j\}_{i\in[n],\, j\in[m]}) = U \otimes V$. To show that the set is linearly independent, assume

$$\sum_{i=1}^{n}\sum_{j=1}^{m} \alpha_{ij} u_i \otimes v_j = 0_{U\otimes V} \tag{1.1}$$

for some coefficients $\alpha_{ij} \in \mathbb{R}$. For some pair $(k,l) \in [n] \times [m]$, define a bilinear map

$$\psi_{kl} : V \times V \to \mathbb{R},\ \psi_{kl}(u_i, v_j) := \delta_{ik}\delta_{jl}.$$

By the universal property of the tensor product, there exists a unique linear map

$$\widehat{\psi}_{kl} : U \otimes V \to \mathbb{R},\ \widehat{\psi}_{kl}(u_i \otimes v_j) = \delta_{ik}\delta_{jl}.$$

Applying $\widehat{\psi}_{kl}$ to the relation (1.1) yields

$$0 = \widehat{\psi}_{kl}\left(\sum_{i=1}^{n}\sum_{j=1}^{m} \alpha_{ij} u_i \otimes v_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{m} \alpha_{ij}\delta_{ik}\delta_{jl} = \alpha_{kl}.$$

Since (k,l) was chosen arbitrary, the set $\{u_i \otimes v_j\}_{i\in[n],\, j\in[m]}$ is linearly independent and thus a basis of $U \otimes V$. □

It follows immediately, that $\dim(U \otimes V) = \dim(U)\dim(V)$, which undermines the rapid growth of dimensionality for higher-order tensor products. Another fairly immediate property of the tensor product is that, it is not commutative in general.

Proposition 1.3 *Let V be a finite-dimensional vector space with $\dim(V) > 1$. Then the tensor product $V \otimes V$ is not commutative.*

Proof. Choose a basis $\{v_i\}_{i\in[n]}$, then by Proposition 1.1 $\{v_i \otimes v_j\}_{i,j\in[n]}$ is a basis of $V \otimes V$, but then $v_1 \otimes v_2 \neq v_2 \otimes v_1$. Otherwise they would be linearly dependent in contradiction to the fact, that they are in the basis of $V \otimes V$. □

Definition 1.4 (Tensor Algebra) Let V be a real vector space. Let $V^{\otimes n}$ denote the n-times iterated tensor product of V, adhering to the convention $V^{\otimes 0} = \mathbb{R}$. The vector space

$$T(V) := \bigoplus_{n=0}^{\infty} V^{\otimes n},$$

is *graded* in the sense that for every $v \in T(V)$ there exists a threshold $m \in \mathbb{N}_0$ such that

$$v = \sum_{n=0}^{m} \pi_n(v)$$

with the canonical projection $\pi_n : T(V) \to V^{\otimes n}$. $T(V)$ becomes the unital, associative *tensor algebra* through elementwise addition and the tensor product extended by bilinearity, in the sense that for every $u, v \in T(V)$

$$u \cdot v = \sum_{i,j} u_i \otimes v_j = \sum_{k\geqslant 0}\sum_{k=i+j} u_i \otimes v_j$$

and the natural action of $\mathbb{R}$ given by $\lambda v = (\lambda v_0, \lambda v_1, \dots) \in T(V)$. Thus, $T(V)$ can be identified with the space of graded sequences with infinitely many zeros, in the sense that

for each sequence element $v_n \in V^{\otimes n}$ [Gre78]. The *extended tensor algebra* is the direct product

$$T((V)) := \prod_{n=0}^{\infty} V^{\otimes n}$$

with elementwise addition and the tensor product by bilinearity in the sense that for every $u, v \in T((V))$

$$u \cdot v = \prod_{k \geqslant 0} \sum_{k=i+j} u_i \otimes v_j.$$

$T((V))$ can be associated with the space of all graded sequences with possibly infinitely many non-zeros [LM22]. $T((V))$ can be endowed with the natural action of $\mathbb{R}$ in the same way as $T(V)$. For $n \in \mathbb{N}$, define the ideal $I_n := \{v \in T((V)) : v_0 = \cdots = v_n = 0\}$. The *truncated tensor algebra* of order n of V is then the quotient algebra

$$T^n(V) := T((V))/I_n \cong \bigoplus_{k=0}^{n} V^{\otimes k}.$$

The canonical homomorphism $T((V)) \to T^n(V)$ is denoted as $\pi_n^{T((V))}$ and similarly, the projection onto the i-th graded element for $i \in [n] \cup \{0\}$ of $T^n(V)$ is denoted as $\pi_i^{T^n(V)} : T^n(V) \to V^{\otimes i}$. It is obvious, that $\iota = (1, 0, 0, \dots)$ is the unit element wrt to the tensor product in both $T(V)$ and $T((V))$.

The inner product is a canonical way to measure similarity of two vectors. In the next subsection, we will formally introduce such a concept for sequences using the signature transform, which requires the notion of an inner product on $T((V))$, given an inner product on V.

Definition 1.5 Let H be a Hilbert space. For $n \in \mathbb{N}$, define an *inner product on $H^{\otimes n}$* through

$$\langle u_{i_1} \otimes \cdots \otimes u_{i_n}, v_{j_1} \otimes \cdots \otimes v_{j_n} \rangle_{H^{\otimes n}} = \prod_{k=1}^{n} \langle u_{i_k}, v_{j_k} \rangle_H.$$

We define an *inner product on the tensor algebra $T((H))$* by setting

$$\langle T_1, T_2 \rangle_{T((H))} = \sum_{n=0}^{\infty} \langle (\pi_n^{T(H)} \circ \pi_n^{T((H))})(T_1), (\pi_n^{T(H)} \circ \pi_n^{T((H))})(T_2) \rangle_{H^{\otimes n}}$$

and similarly on the truncated tensor algebra of order n

$$\langle T_1, T_2 \rangle_{T^n(H)} = \sum_{n=0}^{\infty} \langle \pi_n^{T((H))}(T_1), \pi_n^{T((H))}(T_2) \rangle_{H^{\otimes n}}.$$

For the Signature Transform of a path to exist as sequence of iterated integrals in the extended tensor algebra, some regularity assumptions need to be made.

Definition 1.6 (p-variation) Let V be a Banach space with norm $\|\cdot\|_V$ and $X : [0, s] \to V$ be a continuous path, then for $p \geqslant 1$, and a partition $\mathcal{D} := \{0 \leqslant t_0 < t_1 < \cdots < t_n \leqslant s\}$ of $[0, s]$, the p-variation of x is

$$\|X\|_{p,\,[0,s]} = \left(\sup_{\mathcal{D}} \sum_{i=0}^{n-1} \|x_{i+1} - x_i\|_V^p \right)^{1/p},$$

where the supremum is taken over the set of all finite partitions of $[0, s]$. We denote the collection of continuous paths $X : [0, s] \to V$ with finite p-variation by $\mathscr{V}^p([0, s], V)$.

Definition 1.7 (Young Integral) Let V, W be Banach spaces and $\mathscr{L}(V,W)$ denote the space of linear operators between V and W. For two *paths* $X : [0,s] \to V$ and $Y : [0,s] \to \mathscr{L}(V,W)$ and a partition $\mathcal{D} := \{0 \leqslant t_0 < t_1 < \cdots < t_n \leqslant s\}$ of $[0,s]$, we set $|\mathcal{D}| := \min\{t_{i+1} - t_i : i \in [n-1] \cup \{0\}\}$ and

$$\int_{\mathcal{D}} Y \,\mathrm{d}X = \sum_{i=0}^{n} Y_{t_i}(X_{t_i} - X_{t_{i-1}}),$$

then for arbitrary $t \in [0,t]$ the Young integral of Y along X over the interval $[a;t]$ is given by

$$\int_0^t Y(s)\,\mathrm{d}X(s) := \lim_{j\to\infty} \int_{\mathcal{D}_j} Y \,\mathrm{d}X.$$

where $(\mathcal{D}_j)_{j\in\mathbb{N}}$ is a sequence of partitions of $[0,s]$ such that $|\mathcal{D}_j| \to 0$ as $j \to \infty$ [LM22].

In the setting of the preceding definition, it can be shown that for $p,q \in [1;\infty)$ with $1/p + 1/q > 1$, the Young integral determines a way for a path $Y \in \mathscr{V}^q([0,s],\mathscr{L}(V,W))$ to be integrated along a path $X \in \mathscr{V}^p([0,s],V)$ and yield a well-defined path

$$I : t \mapsto \int_0^t Y(s)\,\mathrm{d}X(s)$$

with finite p-variation, $I \in \mathscr{V}^p([a;b],W)$ [LCL07].

Definition 1.8 (Signature Transform) Let V be a Banach space. The signature transform of a continuous path $X : [0,s] \to V$ is a solution to the *universal differential equation* [LM22]

$$\mathrm{d}S_{0,t}(X) = S_{0,t}(X) \otimes \mathrm{d}X(t),\ S_{0,0}(X) = \iota \in T((V)). \tag{1.2}$$

This already shows that the signature transform is translation-invariant, in a spatial sense, since adding a constant to the path does not alter the solution to the universal differential equation. That this does not hold for time-translations follows from Chen's identity, which will be introduced in the next section. A more tractable format for the signature transform can be written in Young's integration framework.

Theorem 1.9 *Let V be a Banach space and $X \in \mathscr{V}^p([0,s],V)$ for $p \in [1;2)$, then the signature transform of X is given as the path*

$$t \mapsto \prod_{n=0}^{\infty} S^n_{0,t}(X) =: \mathbb{S}_{[0,t]}(X) \in T((V))$$

with $\mathbb{S}_{[0,t]}(X) \in T((V))$ for all $t \in [0,s]$ and

$$S^0_{a,t}(X) := 1,\ S^n_{a,t}(X) := \int_a^t S^{n-1}_{a,s}(X) \otimes \mathrm{d}X(s) \in T^n(V)$$

where the iterated integral is to be taken in the Young sense. Additionally, $\mathbb{S}_{[0,t]}(X)$ is the unique solution to the universal differential equation (1.2). Further, the truncated signature is defined as $\mathbb{S}^n_{[0,t]}(X) := \pi_n^{T((V))} \circ \mathbb{S}_{[0,t]}(X)$.

Proof. We omit the proof of this theorem due to its increased technicality with handling concepts from rough path theory and refer interested readers to Lemma 2.10 of [LCL07]. □

It is now clear that every path $X \in \mathscr{V}^p([0,s],V)$ can be assigned its unique signature transform, however the reverse is not necessarily true, at least for $p \in (1;2)$ this remains an open question [LCL07].

However, for the case $p = 1$ this can be proven, using the concept of *tree-like* equivalence. Loosely speaking, a path is tree-like if one can reduce it to a point by repeatedly erasing back-and-forth pieces. Intuitively, this is exactly like walking on a tree and what happens when every time one comes back along the same branch, that branch cancels. What remains after removing all such backtracks is simply the root of the tree. Tree-like equivalence is in fact a very weak concept, adding an increasing coordinate to a path, e.g. a time-coordinate, already ensures that the original and the transformed path are not tree-like equivalent [KO19].

Let $X, Y \in \mathscr{V}^1([0,s],V)$ be two paths, then $X * Y \in \mathscr{V}^1([0,2s],V)$ is said to be *tree-like* if there exists a continuous *height* function $h : [0,2s] \to [0,\infty)$, such that $h(0) = h(2s) = 0$ and for all $t_1, t_2 \in [2s]$ with $t_1 \leqslant t_2$

$$\|(X * Y)(t_2) - (X * Y)(t_1)\| \leqslant h(t_1) + h(t_2) + \inf_{u \in [0,2s]} h(u).$$

define an equivalence relation $X \sim Y :\Longleftrightarrow X * Y$ tree-like. Theorem 2.29 in [LCL07] shows that $\mathbb{S}_{0,s}(X) = \mathbb{S}_{0,s}(Y)$ iff $X \sim Y$. In the next section, we will make use of the simplex.

1.2 Algebraic Properties

Since the signature transform of a path is unique, defining a signature mapping from the set of paths, modulo translations and reparametrisations, into the tensor algebra is well-defined. In the following, select important algebraical properties of that mapping are defined.

The following statements are only proven for paths with bounded 1*-variation despite being stated for* $p \in [1;2)$*. Proofs for the more general case can be found in [LM22].*

Definition 1.10 (Path Concatenation) Let $X : [0,s] \to V$, $Y : [s,u] \to V$ be two continuous paths. Their concatenation is the path $X * Y : [0,u] \to V$ defined by

$$(X * Y)(t) := \begin{cases} X(t) & \text{for } t \in [0,s] \\ X(s) + Y(t) - Y(s) & \text{for } t \in (s,u]. \end{cases}$$

We will need the notion of a simplex in the following sections.

Definition 1.11 (Simplex) Let $n \in \mathbb{N}$ and $I \subseteq \mathbb{R}$ be a compact interval, then

$$\triangle_n(I) := \{t_1 < \cdots < t_n : t_1, \ldots, t_n \in I\}.$$

The following theorem due to Chen proves that the signature is a homomorphism.

Theorem 1.12 (Chen's Identity) *Let* $X : [0,s] \to V$*,* $Y : [s,u] \to V$ *be two continuous paths with finite* p*-variation for* $p \in [1;2)$*. Then*

$$\mathbb{S}_{[0,u]}(X * Y) = \mathbb{S}_{[0,s]}(X) \otimes \mathbb{S}_{[s,u]}(Y).$$

Proof. Denote the concatenated path as $Z := X * Y$ and fix some truncation level $n \in \mathbb{N}$. There exists a unique index $k \in \mathbb{N}$ such that

$$A_k := \{0 \leqslant t_0 < t_1 < \cdots < t_k < s < t_{k+1} < \cdots \leqslant t_n < u\},$$

then the simplex $\triangle_n(0,u)$ is partitioned by the sets A_k up to a Lebesgue-measure zero set where some $t_i = s$. Since A_k factorizes as a product of simplices, $A_k \cong \triangle_k(0,s) \times \triangle_{n-k}(s,u)$, it follows by Fubini's theorem that,

$$\begin{aligned}
&\int_{A_k} \mathrm{d}Z(t_1) \otimes \cdots \otimes \mathrm{d}Z_{t_n} \\
&= \left(\int_{0<t_1<\cdots<t_k<s} \cdots \int \mathrm{d}X(t_1) \otimes \cdots \otimes \mathrm{d}X(t_k) \right) \otimes \left(\int_{0<t_{k+1}<\cdots<t_n<u} \cdots \int \mathrm{d}Y(t_{k+1}) \otimes \cdots \otimes \mathrm{d}Y(t_n) \right),
\end{aligned}$$

which simply equals $\mathbb{S}^k_{[0,s]}(X) \otimes \mathbb{S}^{n-k}_{[s,u]}(Y)$. Summing over $k \in [n] \cup \{0\}$ yields

$$\mathbb{S}^n_{[0,u]}(Z) = \sum_{k=0}^{n} \mathbb{S}^k_{[0,s]}(X) \otimes \mathbb{S}^{n-k}_{[s,u]}(Y). \qquad \square$$

Additionally, the following inversion property makes the image of the signature a group under path concatenation.

Theorem 1.13 (Inversion Property) *Let $X : [0,s] \to V$ be a continuous path with finite p-variation for $p \in [1;2)$. Define the time-inverted path $\overleftarrow{X} : [s,0] \to V$, $t \mapsto X(s-t)$, then*

$$\mathbb{S}_{[0,s]}(\overleftarrow{X}) = \mathbb{S}_{[0,s]}(X)^{-1}.$$

Proof. Let $t \in [0,s]$. Set $U(t) := 1 - \mathbb{S}_{[0,t]}(X)$. Using the Neumann series, we get that

$$(1+U(t))^{-1} = \sum_{n=0}^{\infty} (-U(t))^n =: \zeta(t),$$

which is well-defined due to the exponential decay of the signature coefficients (Equation 2.52, [LM22]). Differentiating yields

$$\mathrm{d}\zeta(t) = \sum_{n=1}^{\infty} (-1)^n \sum_{k=1}^{n-1} U(t)^k \mathrm{d}U(t) U(t)^{n-1-k}.$$

By the universal differential equation $\mathrm{d}U(t) = (1+U(t)) \otimes \mathrm{d}X(t)$, and thus

$$\begin{aligned}
\mathrm{d}\zeta(t) &= \sum_{n=1}^{\infty} (-1)^n \sum_{k=1}^{n-1} U(t)^k ((1+U(t)) \otimes \mathrm{d}X(t)) U(t)^{n-1-k} \\
&= \left(\sum_{n=1}^{\infty} (-1)^n \sum_{k=0}^{n-1} U(t)^k U(t)^{n-1-k} + \sum_{n=1}^{\infty} (-1)^n \sum_{k=0}^{n-1} U(t)^{k+1} U(t)^{n-1-k} \right) \otimes \mathrm{d}X(t).
\end{aligned}$$

Rearranging and combining gives

$$\mathrm{d}\zeta(t) = \left(\sum_{n=0}^{\infty} (-1)^{n+1} (n+1) U(t)^n \right) (1+U(t)) \otimes \mathrm{d}X(t),$$

but

$$\left(\sum_{n=0}^{\infty}(-1)^{n+1}(n+1)U(t)^n\right)(1+U(t)) =$$
$$\sum_{n=0}^{\infty}(-1)^{n+1}(n+1)U(t)^n + \sum_{n=0}^{\infty}(-1)^{n+1}(n+1)U(t)^{n+1} =$$
$$-1+\sum_{n=1}^{\infty}U(t)^n((-1)^{n+1}(n+1)-(-1)^n n) =$$
$$-1-\sum_{n=1}^{\infty}(-1)^n U(t)^n = -\zeta(t),$$

which yields $\mathrm{d}\zeta(t) = \zeta(t)\otimes \mathrm{d}X(s-t)$ through a change of variables. Since the time-reversed path fulfills $\mathrm{d}\mathbb{S}_{[0,s]}(\overleftarrow{X}) = \mathbb{S}_{[0,s]}(\overleftarrow{X})\otimes \mathrm{d}X(s-t)$ and the solution to the universal differential equation is unique, we conclude $\mathbb{S}_{[0,t]}(X)^{-1} = \zeta(t) = \mathbb{S}_{[0,t]}(\overleftarrow{X})$ for all $t\in[0,s]$. □

Another interesting algebraic property arises from the set of *linear forms* on $T((V))$, i.e. linear functions from $T((V))$ to $\mathbb{R}$.

Let $\{v_1,\dots,v_d\}$ be a basis of V and $\{v_1^*,\dots,v_d^*\}$ be the dual basis of V, i.e. $v_i^*(v_j)=\delta_{ij}$ for all $i,j\in[d]$. Given arbitrary $(\alpha_1,\dots,\alpha_n)\in(V^*)^n$, consider the linear map

$$\varphi_\alpha : V^n \to \mathbb{R}, \varphi_\alpha(u_1,\dots,u_n) = \prod_{i=1}^n \alpha_i(u_i)$$

By the universal property of $V^{\otimes n}$ there exists a unique linear functional $f_\alpha : V^{\otimes n}\to\mathbb{R}$ such that

$$f_\alpha(u_1\otimes\cdots\otimes u_n) = \varphi_\alpha(u_1,\dots,u_n) = \prod_{i=1}^n \alpha_i(u_i)$$

for any $u_1,\dots,u_n\in V$. Note, that the assignment map $(\alpha_1,\dots,\alpha_n)\mapsto f_\alpha\in(V^{\otimes n})^*$ is itself linear and similarly, by the universal property of $(V^*)^{\otimes n}$ there exists a unique linear map $\psi_n : (V^*)^{\otimes n}\to(V^{\otimes n})^*$ such that

$$\psi_n(\alpha_1\otimes\cdots\otimes\alpha_n)(u_1\otimes\cdots\otimes u_n) = f_\alpha(u_1\otimes\cdots\otimes u_n) = \prod_{i=1}^n \alpha_i(u_i).$$

The map ψ_n is an isomorphism since

$$\psi_n(e_{i_1}^*\otimes\cdots\otimes e_{i_n}^*)(e_{j_1}\otimes\cdots\otimes e_{j_n}) = \prod_{k=1}^n \delta_{i_k j_k}$$

and thus ψ_n maps the basis of $(V^*)^{\otimes n}$ to the dual basis of $(V^{\otimes n})^*$. It follows that, $(V^*)^{\otimes n}$ can be canonically identified with $(V^{\otimes n})^*$.

This pairing can be extended to a linear map $(V^{\otimes n})^*\to T((V))^*$ through

$$\widehat{\psi}_n : (V^*)^{\otimes n}\to T((V))^*,\ \widehat{\psi}_n(\xi) = \psi_n(\xi)\circ\pi_n^{T^n(V)}\circ\pi_n^{T((V))}$$

for each $\xi\in(V^*)^{\otimes n}$. Since this is possible for each $n\in\mathbb{N}$, we equally get a graded linear mapping

$$\widehat{\psi} : T(V^*)\to T((V))^*.$$

Thus, for any $n \in \mathbb{N}$ and multi-index $I = (i_1, \dots, i_n) \in [d]^n$, the mapping

$$\widehat{\psi}(e_I^*) \circ \mathbb{S}_{[0,s]}(X) : \mathscr{V}^p([0,s], V) \to \mathbb{R}, X \mapsto \int\limits_{0<t_1<\dots<t_n<s} \cdots \int e_{i_1}^*(\mathrm{d}X(t_1)) \cdots e_{i_n}^*(\mathrm{d}X(t_n))$$

is well-defined. For any two elements of $T(V^*)$ their pointwise product as real functional is a quadratic form on $T((V))$, which in fact endows $T(V^*)$ with an specific algebraic structure.

Definition 1.14 (Shuffle Product) Let $n, m \in \mathbb{N}_{>0}$. The set of (n,m)-*shuffles* is denoted as

$$\mathrm{Sh}(n,m) := \{\sigma \in \mathcal{S}_{n+m} : \sigma(1) < \dots < \sigma(n),\ \sigma(n+1) < \dots < \sigma(n+m)\}$$

where $\mathcal{S}_{n+m}$ is the symmetric group over the set of $n+m$ symbols. Let $e_I^*, e_J^* \in T(V^*)$ with multi-indices $I := (i_1, \dots, i_n) \in [d]^n$, $J := (j_1, \dots, j_m) \in [d]^m$, then their *shuffle product* is defined by

$$e_I^* \sqcup\!\sqcup e_J^* := \sum_{\sigma \in \mathrm{Sh}(n,m)} e^*_{(i_{\sigma^{-1}(1)}, \dots, i_{\sigma^{-1}(n)}, j_{\sigma^{-1}(n+1)}, \dots, j_{\sigma^{-1}(n+m)})}.$$

Intuitively, a shuffle is simply an interleaving of two ordered vectors, that preserves the internal order of each vector.

Theorem 1.15 (Shuffle Property) *Let $e_I^*, e_J^* \in T(V^*)$ with multi-indices $I := (i_1, \dots, i_n) \in [d]^n$, $J := (j_1, \dots, j_m) \in [d]^m$. Then it holds that for all paths $X : [0,s] \to V$ with bounded p-variation for some $p \in [1;2)$,*

$$\widehat{\psi}(e_I^* \sqcup\!\sqcup e_J^*) \circ \mathbb{S}_{[0,s]}(X) = (\widehat{\psi}(e_I^*) \circ \mathbb{S}_{[0,s]}(X))(\widehat{\psi}(e_J^*) \circ \mathbb{S}_{[0,s]}(X)).$$

Proof. Since X is a path of bounded 1-variation, the application of Fubini's theorem is valid and thus the product can be written as

$$\begin{aligned}
&\int_{\triangle_n} e_{i_1}^*(\mathrm{d}X(t_1)) \cdots e_{i_n}^*(\mathrm{d}X(t_n)) \int_{\triangle_m} e_{j_1}^*(\mathrm{d}X(s_1)) \cdots e_{j_m}^*(\mathrm{d}X(s_m)) \\
= &\int_{\triangle_n \times \triangle_m} e_{i_1}^*(\mathrm{d}X(t_1)) \cdots e_{i_n}^*(\mathrm{d}X(t_n))\, e_{j_1}^*(\mathrm{d}X(s_1)) \cdots e_{j_m}^*(\mathrm{d}X(s_m))
\end{aligned}$$

in the Lebesgue sense. Given $(t,s) \in \triangle_n \times \triangle_m$ such that $t_i \neq s_j$ for all $i \in [n], j \in [m]$ there exists a unique $\sigma_{t,s} \in \mathrm{Sh}(n,m)$ such that $\sigma_{t,s}^{-1}(i) < \sigma_{t,s}^{-1}(n+j)$ is equivalent to $t_i < s_j$. For some $\sigma \in \mathrm{Sh}(n,m)$, denote the set $A_\sigma := \{(t,s) \in \triangle_n \times \triangle_m : \sigma = \sigma_{t,s}\}$. Thus,

$$\triangle_n \times \triangle_m = \bigsqcup_{\sigma \in \mathrm{Sh}(n,m)} A_\sigma \cup \bigcup_{i \in [n],\, j \in [m]} \{t_i = s_j\}.$$

Since the second union is a union of hyperplanes of codimension 1, it has measure zero in $\mathbb{R}^{n+m}$. Define the transformation

$$\Phi_\sigma : A_\sigma \to \triangle_{n+m}, \Phi(t,s) = w \text{ with } w := \begin{cases} w_{\sigma(i)} & \text{for } i \in [n] \\ w_{\sigma(n+j)} & \text{for } j \in [m], \end{cases}$$

then $|\det(\mathrm{D}\Phi_\sigma^{-1})| = 1$ since the inverse of the Jacobian is simply a permutation matrix. We conclude

$$\begin{aligned}
&\int_{\triangle_n \times \triangle_m} e_{i_1}^*(\mathrm{d}X(t_1)) \cdots e_{i_n}^*(\mathrm{d}X(t_n))\, e_{j_1}^*(\mathrm{d}X(s_1)) \cdots e_{j_m}^*(\mathrm{d}X(s_m)) \\
= &\sum_{\sigma \in \mathrm{Sh}(n,m)} \int_{\triangle_{n+m}} e^*_{k_{\sigma^{-1}(1)}}(\mathrm{d}X(w_1)) \cdots e^*_{k_{\sigma^{-1}(n+m)}}(\mathrm{d}X(w_{n+m})).
\end{aligned}$$

□

We note, that the result extends to all $v^*, w^* \in T(V^*)$ due to linearity. In fact, Theorem 1.15 states the fact, that the linear forms on $T((V))$ induced by $T(V^*)$ restricted to the image of the signature form an algebra of real-valued functions [LCL07].
The shuffle property shows that the signature can be seen as the path-space generalization of the monomial feature map. In $\mathbb{R}^d$, the application of linear functionals equals iterated integrals of the coordinate increments, which plays the role of non-commutative monomials with degree being the level of the dual basis elements. The shuffle property, is then the analogue of multiplying monomials and collecting terms.

1.3 Kernel Methods

A canonical approach to learning a functional relationship $f : X \to Y$ given a finite dataset $\Omega \subseteq X \times Y$ is to define basic features and fit a linear combination of these. In polynomial regression, these features are monomials evaluated at each $(x, y) \in \Omega$ up to a certain fixed degree, chosen as hyperparameter.

Transferring this idea to sequentially ordered data, is in fact possible, however we will first introduce some notation. Let V be a Banach space. Denote

$$\mathscr{X}_V := \bigcup_{n=1}^{\infty} \{(x_1, \dots, x_n) \in V^n\}$$

the collection of all sequences of finite but arbitrary length over V. In this supervised learning setting, consider a labelled dataset $\Omega := \{(x_i, y_i)\} \subseteq \mathscr{X}_V \times \mathbb{R}$ of size $n \in \mathbb{N}_{>0}$ and denote $\pi_{\mathscr{X}_V}$ the projection onto $\mathscr{X}_V$. The task is to determine an estimator of a function $f : \mathscr{X}_V \to \mathbb{R}$ given Ω, that is capable of predicting the response to new inputs. Theoretical guarantees on the error of such approximations can be made when features are extracted using a *universal feature map* Ψ [LM22]. It provides a lifted representation of the input features $\pi_{\mathscr{X}_V}(\Omega)$ as a subset of a Banach space W, in which continuous functions can be approximated by linear functions arbitrarily well. Equivalently, in mathematical terms, for any $g \in \mathscr{C}^0(\Psi(\pi_{\mathscr{X}_V}(\Omega)))$ and $\varepsilon > 0$ there exists a real linear functional $\xi \in \Psi(\pi_{\mathscr{X}_V}(\Omega))^*$ such that

$$\sup_{w \in \Psi(\pi_{\mathscr{X}_V}(\Omega))} |f(w) - \xi(w)| \leqslant \varepsilon.$$

Given some convex loss function $\ell : \mathbb{R} \times \mathbb{R} \to [0, \infty)$, pulling back along Ψ results in the following convex optimization problem for empirical risk minimization

$$\min_{\xi \in \Psi(\pi_{\mathscr{X}_V}(\Omega))^*} \sum_{i=1}^{n} \ell(\xi(\Psi(x_i)), y_i) + \lambda \|\xi\|_{W^*}^2 \tag{1.3}$$

for some regularization parameter $\lambda > 0$ and the operator norm $\|\xi\|_{W^*} = \sup\{\xi(w) : w \in W, \|w\|_W \leqslant 1\}$.

Choosing W in this situation such that it is a Hilbert space, has two important advantages. If W is a Hilbert space, then by the Fréchet-Riesz representation theorem [Cla20], W^* is isometrically isomorphic to W and for any fixed $x \in \pi_{\mathscr{X}_V}(\Omega)$ the function

$$\mathscr{k} : \pi_{\mathscr{X}_V}(\Omega) \to \mathbb{R}, \ \mathscr{k}(x, x_i) := \langle \Psi(x), \Psi(x_i) \rangle_W$$

is positive definite and symmetric, i.e. a kernel. Kernel functions can be informally regarded as similarity measures, which appear as such in various statistical learning algorithms, such

as support vector regression or principal component analysis. It generates the Reproducing Kernel Hilbert Space,

$$\mathscr{H}_{\mathscr{k}} = \overline{\operatorname{span}\{\mathscr{k}(x,\cdot) : x \in \pi_{\mathscr{X}_V}(\Omega)\}},$$

containing the images of the input patterns under Ψ, or in more rigorous terms, the Reproducing Kernel Hilbert Space can be defined as a Hilbert space of functions g on $\pi_{\mathscr{X}_V}(\Omega)$ such that all evaluation functionals $g \mapsto g(x)$, $x \in \pi_{\mathscr{X}_V}(\Omega)$ are continuous [SS01]. In $\mathscr{H}_{\mathscr{k}}$ each element $f \in \mathscr{H}_{\mathscr{k}}$ is represented by a unique element $\mathscr{k}(x,\cdot)$ such that

$$f(x) = \langle f, \mathscr{k}(x,\cdot)\rangle_{\mathscr{H}_{\mathscr{k}}}.$$

One advantage is, that the representer theorem [SHS01] ensures that a minimizer for (1.3) can be written as a linear combination in the kernel sections $\mathscr{k}(x,\cdot)$, which reduces the optimization problem to a finite-dimensional one. The second, more important advantage, often referred to as kernel trick, is that evaluating such a minimizer and in general any function in the Reproducing Kernel Hilbert Space, requires only computing the Gram matrix $K_{ij} := (\mathscr{k}(x_i, x_j))_{i,j\in[n]}$, i.e. the values of the kernel for each input pair $(x_i, x_j) \in \mathscr{X}_V \times \mathscr{X}_V$, since

$$f(x) = \sum_{i=1}^{n} \alpha_i \mathscr{k}(x, x_i). \tag{1.4}$$

In conclusion, kernel methods enable learning non-linear relationships through linear models in a lifted feature space.

With regard to the signature, we mentioned that it can be seen in close analogy to monomials on the domain of sequentially ordered data, and is, in accordance with the polynomial intuition, indeed a valid universal feature map according to Theorem 5.4 in [CST23]. This gives rise to the following

Definition 1.16 (Signature Kernel) Let H be a Hilbert space, $I \subseteq \mathbb{R}$ be a compact interval, then

$$\mathscr{k}_{\mathtt{Sig}} : \mathscr{V}^p(I,H) \times \mathscr{V}^p(I,H) \to [0;\infty), (X,Y) \mapsto \langle \mathbb{S}_I(X), \mathbb{S}_I(Y)\rangle_{T((H))} \tag{1.5}$$

defines a positive-semidefinite kernel, where the inner product was introduced in Definition 1.5. Similarly, the truncated signature kernel of order n is defined as

$$\mathscr{k}^n_{\mathtt{Sig}}(X,Y) := \langle (\pi_n^{T((H))} \circ \mathbb{S}_I)(X), (\pi_n^{T((H))} \circ \mathbb{S}_I)(Y)\rangle_{T^n(H)}. \tag{1.6}$$

Theorem 1.17 *Let $p \in [1;2)$. Then the functions defined in (1.5) and (1.6) are positive-semidefinite kernels.*

Proof. By Theorem 3.7 in [LCL07] and the Cauchy-Schwarz inequality, it holds that,

$$\begin{aligned}
|\mathscr{k}_{\mathtt{Sig}}(X,Y)| &= \sum_{n=0}^{\infty} |\langle (\pi_n^{T((H))} \circ \mathbb{S}_I)(X), (\pi_n^{T((H))} \circ \mathbb{S}_I)(Y)\rangle_{T^n(H)}| \\
&\leqslant \sum_{n=0}^{\infty} \|\mathbb{S}_I^n(X)\|_{T^n(H)} \|\mathbb{S}_I^n(Y)\|_{T^n(H)} \\
&\leqslant \sum_{n=0}^{\infty} \frac{(\|X\|_{p,I}\|Y\|_{p,I})^{n/p}}{\Gamma(C_p+1)^2} \leqslant E_{0,\Gamma(C_p+1)}(\|X\|_{p,I}) E_{0,\Gamma(C_p+1)}(\|Y\|_{p,I}) < \infty
\end{aligned}$$

with some constant C_p that only depends on p, defined in Theorem 3.7 of [LCL07] and the Mittag-Leffler function $E_{0,\Gamma(C_p+1)}$ as in Equation 1.2 of [HMS11], thus the kernels are well-defined. Symmetry and positive-semidefiniteness are obvious since both the signature kernel and the truncated signature kernel are inner products of their respective feature maps. □

One major practical issue with the kernels introduced above is that sequences are most certainly not observed as continuous paths but rather recorded in a finite amount of *discrete timestamps* or increments. Under this assumption, [KO19] suggest using the following approximation,

$$\sum_{1\leqslant i_1<\cdots<i_n\leqslant n}\cdots\sum \Delta X(t_{i_1})\otimes\cdots\otimes\Delta X(t_{i_n}) \approx \int_{\triangle_n(I)} \mathrm{d}X(t_1)\otimes\cdots\otimes\mathrm{d}X(t_n) \tag{1.7}$$

where $\Delta X(t_{i_j}) := X(t_{i_j}) - X(t_{i_{j-1}})$ for every $j \in [n]$ are the first-order differences of X. Using this approximation is in fact justified, which was shown in Theorem 5 of [KO19]. Thus the approximation in (1.7) can be used in the following *discretized version* of the signature kernel and likewise the truncated signature kernel.

Definition 1.18 (Discretized Signature Kernel) Let H be a Hilbert space, $I \subseteq \mathbb{R}$ be a compact interval and denote the length or number of elements of a sequence X by $|X|$, then the *discretized signature kernel* is denoted as

$$\widetilde{\mathscr{k}}_{\mathtt{Sig}}(X,Y) := \sum_{n=0}^{\infty} \sum_{\substack{i\in\triangle_p([|X|])\\ j\in\triangle_p([|Y|])}} \prod_{p=1}^{n} \langle \Delta(X(t_i)), \Delta Y(t_j)\rangle_H.$$

Similarly, the *truncated discretized signature kernel* of order n is defined as

$$\widetilde{\mathscr{k}}^{n}_{\mathtt{Sig}}(X,Y) := \sum_{k=0}^{n} \sum_{\substack{i\in\triangle_p([|X|])\\ j\in\triangle_p([|Y|])}} \prod_{p=1}^{k} \langle \Delta(X(t_i)), \Delta Y(t_j)\rangle_H.$$

Note that in the discretized setting, for a discrete path X with samples at times $t_0, \ldots, t_n$, the p-variation norm from Definition 1.6 reduces to

$$\|X\|_p = \left(\sum_{i=1}^{n} \|\Delta X(t_i)\|^p\right)^{1/p}.$$

Chapter 2
Dimensionality Reduction

In the previous chapter, it became clear that the tensor structure of the signature transform is a fruitful framework for linear regression with theoretical performance guarantees, however a significant drawback is the rapidly increasing dimensionality of the truncated signature as seen in Proposition 1.2. In fact, for the truncated signature of order $n \in \mathbb{N}_{>0}$ and some path $X \in \mathscr{V}^p([0,s], H)$ with $p \in [1;2)$ for some finite-dimensional Hilbert space H,

$$\dim(\pi_n^{T((H))} \circ \mathbb{S}_{0,s}(X)) = \sum_{k=0}^{n} \dim(H^{\otimes k}) = \begin{cases} \frac{\dim(H)^{n-1}-1}{\dim(H)-1} & \text{for } \dim(H) \geqslant 2; \\ n+1 & \text{for } \dim(H) = 1. \end{cases}$$

As outlined in the previous section, kernel methods circumvent this issue with the kernel trick. However, they require the computation of the Gram matrix. Even for a dataset of moderate size this could become an issue since it scales quadratically in the number of sequences n or more precisely as $\mathscr{O}(n^2 L_{\max}^2 \dim(H))$ where $L_{\max}$ [TOS25] is the maximum length of a sequence. In the following chapter, we analyse two prominent dimensionality reduction techniques to achieve linear complexity in the number of sequences.

2.1 Random Fourier Features

To extend learning with shift-invariant kernels, i.e. kernels that are invariant under common translations, to large-scaled datasets, the work of [RR07] avoids relying on the kernel trick and explicitly computing the Gram matrix but instead approximates the kernel function directly using probabilistic techniques. They project the data to a low-dimensional Euclidean inner product space using a randomized feature map $z : H \to \mathbb{R}^d$ so that the inner product between a pair of transformed points is an unbiased estimator of the true kernel function. The image under z is low-dimensional and thus enables using fast linear learning methods for training but also for evaluating; using (1.4) requires $\mathscr{O}(n \dim(H))$, however using the learned hyperplane after applying the random feature map only requires a complexity of $\mathscr{O}(\dim(H) + d)$ in time and storage [RR07]. The derivation of a possible random feature map, that satisfies the assumption of unbiasedness, is based on the following theorem, which is due to [Rud90].

Theorem 2.1 (Bochner's Theorem) *A continuous, shift-invariant kernel $\mathscr{k} : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ is positive-definite iff $\mathscr{k}$ is the Fourier transform of a non-negative, finite measure Λ,*

$$\mathscr{k}(x,y) = \int_{\mathbb{R}^d} \exp(\mathrm{i}\langle w, x-y\rangle)\, \mathrm{d}\Lambda(w).$$

Λ is often referred to as spectral measure of $\mathscr{k}$.

An immediate consequence is that when scaling the kernel as $\mathscr{k}'(x,y) = \mathscr{k}(x,y)/\mathscr{k}(x,x)$, the spectral measure of $\mathscr{k}'$ becomes a probability measure since $\Lambda(\mathbb{R}^d) = \mathscr{k}(x,x) < \infty$. Setting $\zeta_w(x) = \exp(\mathrm{i}\langle w, x\rangle)$, yields

$$\mathscr{k}'(x,y) = \frac{1}{\Lambda(\mathbb{R}^d)} \int_{\mathbb{R}^d} \zeta_w(x)\overline{\zeta_w(y)}\, \mathrm{d}\Lambda(w) = \mathbb{E}[\zeta_w(x)\overline{\zeta_w(y)}],$$

where $\overline{\zeta_w(y)}$ denotes the complex conjugate and the expectation is taken wrt the probability measure $\Lambda(\cdot)/\Lambda(\mathbb{R}^d)$. Since $\mathscr{k}$ is real and subsequently also Λ, the complex exponential ζ_w reduces to the cosine. This gives rise to the following

Theorem 2.2 (Random Fourier Features) *Let $\mathscr{k} : \mathbb{R}^d \times \mathbb{R}^d \to [0; \infty)$ be a shift-invariant, continuous and positive-definite kernel and $\beta \sim \mathrm{Uniform}(0, 2\pi)$ be a random offset. Assume wlog that $\mathscr{k}$ is properly scaled such that its spectral measure Λ is also a probability measure. Draw n samples $w_1, \ldots, w_n \sim \Lambda$ and define the feature map*

$$z_n(x) = \sqrt{\frac{2}{d}}(\cos(\langle w_1, x\rangle + \beta), \ldots, \cos(\langle w_n, x\rangle + \beta))^\intercal,$$

then

$$\mathscr{k}(x, y) = \mathbb{E}[\langle z_n(x), z_n(y)\rangle].$$

Proof. Note that this follows pretty much by construction of the feature map. Let $(z_n(x))_i$ be the i-th component of $z_n(x)$. Then, for all $i \in [n]$, using basic identities from trigonometry, it holds that

$$\begin{aligned}
&\mathbb{E}_\beta[(z_n(x))_i (z_n(y))_i] \\
&= \mathbb{E}_\beta[\cos(\langle w_i, x\rangle + \beta)\cos(\langle w_i, y\rangle + \beta)] \\
&= \frac{1}{2}\,\mathbb{E}_\beta[\cos(\langle w_i, x - y\rangle) + \cos(\langle w_i, x + y\rangle) + 2\beta] \\
&= \frac{1}{2}(\cos(\langle w_i, x - y\rangle) + (\cos(\langle w_i, x + y\rangle)\mathbb{E}_\beta[\cos(2\beta)] - \sin(\langle w_i, x + y\rangle)\mathbb{E}_\beta[\sin(2\beta)])) \\
&= \frac{1}{2}(\cos(\langle w_i, x - y\rangle),
\end{aligned}$$

which already concludes the proof by linearity of expectation. □

The Random Fourier Signature kernel is thus simply a Monte-Carlo method for approximating the true kernel. The following theorem, concludes this introduction and shows that the feature map in Theorem 2.2 estimates the true kernel globally on compact sets $K \subseteq \mathbb{R}^d$ with arbitrary precision $\varepsilon > 0$ [RR07] given the sample size

$$n = \Omega\left(\frac{d}{\varepsilon^2}\ln\left(\frac{\sigma_\Lambda^2\,\mathrm{diam}(K)}{\varepsilon}\right)\right).$$

Theorem 2.3 (Uniform Convergence of Random Fourier Features [RR07]) *Let $K \subseteq \mathbb{R}^d$ be a compact set and $\mathscr{k} : \mathbb{R}^d \times \mathbb{R}^d \to [0; \infty)$ a continuous, translation-invariant kernel such that its spectral measure Λ has finite second moment $\sigma_\Lambda^2 < \infty$. Then, for samples $w_1, \ldots, w_n \sim \Lambda$ iid and $\varepsilon > 0$,*

$$\mathbb{P}\left(\sup_{x,y\in K} |\langle z_n(x), z_n(y)\rangle - \mathscr{k}(x, y)| \geqslant \varepsilon\right) \leqslant C_d\left(\frac{2\sigma_\Lambda^2\,\mathrm{diam}(K)}{\varepsilon}\right)^{2d/(d+2)} \exp\left(-\frac{n\varepsilon^2}{4(d+2)}\right)$$

where z_n is the random mapping as in Theorem 2.2 and $C_d > 0$ is a constant that only depends on d.

To prove the theorem, we need to introduce the following notions from combinatorics in metric spaces and two important concentration inequalities.

Definition 2.4 Let $\varepsilon > 0$, (T, d) be a metric space and A be a non-empty subset of T.

A set $N_\varepsilon \subseteq A$ is said to be a ε-*net* for A wrt d if for every point $x \in A$, there exists a point $\nu \in N_\varepsilon$ such that $d(x, \nu) < \varepsilon$ [Cle63].

Theorem 2.5 (Markov's Inequality) *Let X be a non-negative, integrable random variable and $\varepsilon > 0$, then*

$$\mathbb{P}(X \geqslant \varepsilon) \leqslant \frac{\mathbb{E}[X]}{\varepsilon}.$$

Proof. For each $\varepsilon > 0$, it holds that $X \geqslant \varepsilon \mathbb{1}(\{X \geqslant \varepsilon\})$, and by the monotonicity of the expectation, we conclude

$$\mathbb{E}[X] \geqslant \mathbb{E}[\varepsilon \mathbb{1}(\{X \geqslant \varepsilon\})] = \varepsilon \mathbb{P}(X \geqslant \varepsilon). \qquad \square$$

Theorem 2.6 (Hoeffding's Inequality) *Let $X_1, \ldots, X_n$ be independent random variables such that $X_i \in [\alpha_i, \beta_i]$ a.s. Then for any $\varepsilon > 0$,*

$$\mathbb{P}\left(\sum_{i=1}^{n}(X_i - \mathbb{E}[X_i]) \geqslant \varepsilon\right) \leqslant \exp\left(-\frac{2\varepsilon^2}{\sum_{i=1}^{n}(\beta_i - \alpha_i)^2}\right)$$

Proof. See Theorem 2.2.1 in [Ver18]. $\square$

Proof of Theorem 2.3 (due to [RR07]). Since K is compact, $\operatorname{diam}(K) < \infty$, thus one can define $K_\Delta := \{x - y : x, y \in K\} \subseteq B(0, 2\operatorname{diam}(K))$. Further, because $\mathscr{k}$ is translation-invariant, one can write $\mathscr{k}(x, y) = \mathscr{k}_\Delta(x - y)$ for some $\mathscr{k}_\Delta : \mathbb{R}^d \to \mathbb{R}$ and all $x, y \in K$ wlog. Hence,

$$\sup_{x,y \in K} |\langle z_n(x), z_n(y)\rangle - \mathscr{k}(x, y)| = \sup_{\delta \in K_\Delta} \underbrace{\left|\langle z_n(0), z_n(\delta)\rangle - \mathscr{k}_\Delta(\delta)\right|}_{r(\delta)},$$

which justifies proving the uniform bound over K_Δ.

Since K_Δ is compact, for $\eta > 0$, there exists a finite η-net N_η over K_Δ [Cle63]. Since $\operatorname{diam}(K_\Delta) \leqslant 2\operatorname{diam}(K)$ a standard volumetric bound [CS01] yields

$$|N_\eta| \leqslant \left(\frac{4\operatorname{diam}(K)}{\eta}\right)^d. \tag{2.1}$$

By differentiating one obtains

$$\nabla\langle z_n(0), z_n(\delta)\rangle = -\frac{1}{n}\sum_{i=1}^{n} w_i \sin(\langle w_i, \delta\rangle)$$

and thus $\mathbb{E}[\nabla r(\delta)] = 0$. Now it follows that,

$$\mathbb{E}[\|\nabla r(\delta)\|_2^2] \leqslant \frac{\sigma_\Lambda^2}{n} \tag{2.2}$$

for all $\delta \in K_\Delta$. Let $\delta^* := \arg\max\{\|\nabla r(\delta)\|_2 : \delta \in K_\Delta\}$ and define the random Lipschitz constant $L_\Lambda := \|\nabla r(\delta^*)\|_2$. By Markov's inequality and (2.2),

$$\mathbb{P}\left(L_\Lambda \geqslant \frac{\varepsilon}{2\eta}\right) = \mathbb{P}\left(L_\Lambda^2 \geqslant \frac{\varepsilon^2}{4\eta^2}\right) \leqslant \frac{\mathbb{E}[\|\nabla r(\delta^*)\|_2^2]}{\varepsilon^2(4\eta^2)} \leqslant \left(\frac{2\eta\sigma_\Lambda}{\varepsilon n}\right)^2.$$

Let $\nu \in N_\eta$ be an arbitrary net-anchor. Then by Hoeffding's inequality,

$$\mathbb{P}\left(|\langle z_n(0), z_n(\nu)\rangle - \mathbb{E}[\langle z_n(0), z_n(\nu)\rangle]| \geqslant \varepsilon\right) = \mathbb{P}\left(|r(\nu)| \geqslant \varepsilon\right) \leqslant 2\exp\left(-\frac{n\varepsilon^2}{8}\right)$$

since $|\langle z_n(0), z_n(\nu)\rangle| = 1$. By definition, for any $\delta \in K_\Delta$ there exists a net-anchor $\nu_\delta \in N_\eta$ such that $\|\delta - \nu_\delta\|_2 \leqslant \eta$. On the event

$$\mathscr{E} := \left\{ L_\Lambda \leqslant \frac{\varepsilon}{2\eta} \right\} \cap \left\{ \max_{\nu \in N_\eta} |r(\nu)| \leqslant \frac{\varepsilon}{2} \right\}$$

it holds that,

$$\begin{aligned} |r(\delta)| &\leqslant |r(\nu_\delta) - r(\delta)| + |r(\nu_\delta)| \\ &\leqslant L_\Lambda \|\nu_\delta - \delta\|_2 + \max_{\nu \in N_\eta} |r(\nu)| \leqslant \varepsilon \end{aligned}$$

for any $\delta \in K_\Delta$ by the Lipschitz-continuity of r . Thus, to show uniform concentration it remains to bound the complementary event $\mathscr{E}^\complement$ and show that it occurs with sufficiently low probability. Using a union bound over all anchors $\nu \in N_\eta$ and (2.1) it follows that,

$$\begin{aligned} \mathbb{P}\left(\sup_{\delta \in K_\Delta} |r(\delta)| \geqslant \varepsilon \right) &\leqslant \inf_{\eta > 0} \left\{ \mathbb{P}\left(L_\Lambda \geqslant \frac{\varepsilon}{2\eta} \right) + \mathbb{P}\left(\bigcup_{\nu \in N_\eta} \left\{ r(\nu) \geqslant \frac{\varepsilon}{2} \right\} \right) \right\} \\ &\leqslant \inf_{\eta > 0} \left\{ \left(\frac{2\eta}{\varepsilon n} \right)^2 \sigma_\Lambda^2 + 2|N_\eta| \exp\left(-\frac{2n\varepsilon^2}{8} \right) \right\} \\ &\leqslant \inf_{\eta > 0} \left\{ \underbrace{\left(\frac{2}{\varepsilon n} \right)^2 \sigma_\Lambda^2}_{c_1} \eta^2 + \underbrace{\left(2^{2+1/d} \operatorname{diam}(K) \right)^d \exp\left(-\frac{n\varepsilon^2}{8} \right)}_{c_2} \eta^{-d} \right\}. \end{aligned}$$

Taking the derivative of $g(\eta) := c_1 \eta^2 + c_2 \eta^{-d}$ yields the optimal solution

$$\eta^* = \left(\frac{d c_1}{2 c_2} \right)^{1/(d+2)}$$

By plugging η^* back into (2.1) one obtains the final result,

$$\mathbb{P}\left(\sup_{\delta \in K_\Delta} |r(\delta)| \geqslant \varepsilon \right) \leqslant C_d \left(\frac{\sigma_\Lambda^2 \operatorname{diam}(K)}{\varepsilon} \right)^{2d/(d+2)} \exp\left(-\frac{n\varepsilon^2}{4(d+2)} \right)$$

where

$$C_d := 2^{(6d+2)/(d+2)} \left(\left(\frac{d}{2} \right)^{2/(d+2)} + \left(\frac{2}{d} \right)^{d/(d+2)} \right) > 0. \qquad \square$$

As mentioned previously, the complexity of the signature kernel scales poorly in the number of sequences. One possible way to enjoy the benefit of linear sequence length and low-dimensional feature complexity with theoretical guarantees is through Random Fourier Features [TOS25].

One particular issue worth mentioning is that a direct application of the standard Random Fourier Features as derived by Bochner's theorem does not apply since the space of all sequences of bounded p-variation is not a linear space, however uniform bounds can still be derived using concentration inequalities and tail bounds similar to the Random Fourier Feature construction [TOS25]. Assuming a.s. boundedness of the spectral measure in that case is a very restrictive assumption, whereas properties such as existence of the moment-generating function, tail bounds and moment-boundedness are sufficient for the statements to hold.

Definition 2.7 (Bernstein Moment Condition) Let Λ be a d-dimensional probability measure and take an arbitrary sample $w \in \mathbb{R}^d$, then it is called (α^2, β)-Bernstein if there exist constants $\alpha^2, \beta > 0$ such that

$$\mathbb{E}[w_i^{2n}] \leqslant \frac{n!\alpha^2\beta^{n-2}}{2}$$

for all $i \in [d]$, $n \in \mathbb{N}$.

Definition 2.8 (Random Fourier Signature Features) Fix some truncation level $m \in \mathbb{N}$ and a translation-invariant, continuous, positive-definite kernel $\mathscr{k} : \mathbb{R}^d \times \mathbb{R}^d \to (0; \infty)$ with (α^2, β)-Bernstein spectral measure Λ. Let $W^{(1)}, \dots, W^{(m)} \in \mathbb{R}^{d \times n}$ be Random Fourier Features, with each matrix $W^{(i)} \sim \Lambda$ containing n iid samples for each $i \in [m]$. Define the random feature map

$$\widetilde{\varphi}_i(X) := \frac{1}{\sqrt{n}}(\cos(\langle W^{(i)}, X\rangle)), \sin(\langle W^{(i)}, X\rangle))^\intercal \in \mathbb{R}^{2n} \tag{2.3}$$

for each $i \in [m]$ and the corresponding tensorized feature map

$$\widetilde{\varphi}_{\mathtt{RFF},m}(X) := \sum_{k=0}^{m} \sum_{i \in \Delta_k([|X|])} \Delta\widetilde{\varphi}_1(X(t_{i_1}) \otimes \cdots \otimes \Delta\widetilde{\varphi}_m(X(t_{i_k})$$

for all sequences $X \in \mathscr{X}_{\mathbb{R}^d}$ with the usual first-order difference $\Delta\widetilde{\varphi}_i(X(t_{i_j})) = \widetilde{\varphi}_i(X(t_{i_j})) - \widetilde{\varphi}_i(X(t_{i_{j-1}}))$. Then the truncated *Random Fourier Signature Feature* kernel of order m is defined as

$$\widetilde{\mathscr{k}}^m_{\mathtt{RFF}}(X, Y) := \langle \widetilde{\varphi}_{\mathtt{RFF},m}(X), \widetilde{\varphi}_{\mathtt{RFF},m}(Y)\rangle_{T^m(\mathbb{R}^d)}.$$

Note that the authors chose a different Random Fourier Feature Map, however this is still in line with the original construction from Theorem 2.2 due to the trigonometric identity $\cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta) = \cos(\alpha - \beta)$, thus the original shape, omitting the random shift sampled uniformly from $[0; 2\pi]$, is recovered by the bilinearity of the inner product. The discretized signature kernel is thus prepended by the Random Fourier Feature Map of some translation-invariant, continuous and positive-definite kernel $\mathscr{k} : \mathbb{R}^d \times \mathbb{R}^d \to (0; \infty)$, e.g. the RBF kernel [TOS25], to achieve considerable dimensionality reduction by mapping each sequence to a explicit finite-dimensional vector and skipping all pairwise interactions between sequences that are left to compute. The following theorem guarantees that the truncated discretized signature kernel can be approximated arbitrarily precise on the ball of sequences over compact state-spaces $K \subseteq \mathbb{R}^d$ and 1-variation bounded by some constant.

Theorem 2.9 (Uniform Convergence of Random Fourier Signature Features) *Let $m \in \mathbb{N}$, $V > 0$ and $\mathscr{k} : \mathbb{R}^d \times \mathbb{R}^d \to [0; \infty)$ be a continuous, positive-semidefinite and shift-invariant kernel with (α^2, β)-Bernstein spectral measure Λ. Let $K \subseteq \mathbb{R}^d$ be a compact set, then we denote the set of all sequences with p-variation bounded by V as*

$$B_{\|\cdot\|_p, K}(0, V) = \{X \in \mathscr{X}_K : \|X\|_p \leqslant V\}.$$

Then we have that for all $\varepsilon > 0$,

$$\begin{aligned}
&\mathbb{P}\left(\sup_{X,Y \in B_{\|\cdot\|_1, K}(0,V)} |\widetilde{\mathscr{k}}^m_{\mathtt{Sig}}(X, Y) - \widetilde{\mathscr{k}}^m_{\mathtt{RFF}}(X, Y)| \geqslant \varepsilon\right) \\
&\leqslant m\theta_{n,d,I}\left(\left(\frac{\tau_{d,m,V}}{\varepsilon}\right)^{d/(d+1)} + d + 1\right)\exp\left(-\xi_{n,d,\Lambda}\min\left\{\left(\frac{\varepsilon}{\tau_{d,m,V}}\right)^2, \left(\frac{\varepsilon}{\tau_{d,m,V}}\right)^{1/m}\right\}\right),
\end{aligned}$$

with the constants defined as

$$\tau_{d,m,V} := m(2V^2 \max\{\|\mathbb{E}[ww^\intercal]\|_2^{1/2}, 1\} \max\{\sigma_\Lambda^2, d\})^m;$$
$$\xi_{n,d,\Lambda} := \frac{n}{2(d+1)(\alpha^2+\beta)};$$
$$\theta_{n,d,I} := 2^{\frac{4(d+1)+1}{d+1}} \operatorname{diam}(K)^{\frac{d}{d+1}} \sum_{i,j=1}^{d} \left(\sup_{x,y\in K} \left\| \nabla \left(\frac{\partial \mathscr{k}(x,y)}{\partial x \partial y} \right) \right\|_2 + \mathbb{E}[w_i w_j \|w\|_2] \right)^{d/(d+1)}.$$

Proof. See Theorem C.10 in [TOS25]. □

The Random Fourier Signature Feature kernel enjoys linear complexity in the number of sequences N, or more precisely $\mathscr{O}(NL_{\max}(mnd + 1 + n + \cdots + n^m))$, however still admits a tensor structure and thus polynomial complexity in the number of Random Fourier Feature samples n [TOS25].

2.2 Tensor Random Projections

Random projections replace the original high-dimensional dataset $D \in \mathbb{R}^{n\times d}$ by a linear subspace through the origin using a multiplication with a random matrix $A \in \mathbb{R}^{d\times k}$ where $k \ll \min\{n, d\}$ [BM01]. Despite its name suggesting orthogonal columns, this does not need to be assumed for A, in fact it suffices if its columns have unit length. The idea of such projections is rooted in the Johnson-Lindenstrauss-Lemma, which states that if the dimension, which is projected onto, is chosen as $\mathscr{O}(\varepsilon^{-2}\ln(n))$, *all* pairwise Euclidean distances and inner products are preserved up to precision $(1 \pm \varepsilon)$ for any $\varepsilon \in (0; 1)$ [DG03]. Usually the random matrix entries are chosen to be iid Gaussian, however all zero-mean, unit-variance distributions could be applied in this framework [BM01]. Low memory versions of random projections were introduced by [Ach01] and subsequently the computationally more efficient *very-sparse random projection* followed, which was developed by [LHC06] for fast querying in database applications.

Definition 2.10 (Very-Sparse Random Projection) Let $D \in \mathbb{R}^{n\times d}$ be a dataset. The *very-sparse random projection matrix* $\Pi \in \mathbb{R}^{d\times k}$ is based on the sparse projection matrix by [Ach01] and defined as

$$\Pi_{ij} := \begin{cases} -\sqrt{\theta} & \text{with probability } 1/(2\theta) \\ 0 & \text{with probability } 1 - 1/\theta \\ \sqrt{\theta} & \text{with probability } 1/(2\theta) \end{cases}$$

for some sparsity parameter θ that can be chosen as $\theta \in [1, \sqrt{d}]$.

Tensor random projections extend this to tensor features. In alignment with Proposition 1.2, the tensor products' dimensions grow rapidly. To reduce storage requirements for the equally large random projections, [Sun+21] suggest using the matching column-wise Kronecker product, or Khatri-Rao product. For two matrices $A \in \mathbb{R}^{n\times k}$, $B \in \mathbb{R}^{m\times k}$, their *Khatri-Rao product* is defined as

$$A \odot B := \begin{pmatrix} A_{11}B_{*,1} & \cdots & A_{1k}B_{*,k} \\ \vdots & \ddots & \vdots \\ A_{n1}B_{*,1} & \cdots & A_{nk}B_{*,k} \end{pmatrix} \in \mathbb{R}^{nm\times k}.$$

Definition 2.11 (Tensor Random Projection) Let $T \in (\mathbb{R}^d)^{\otimes n}$ and take arbitrary random projections $A_i \in \mathbb{R}^{d\times k}$ for each $i \in [n]$, then the *tensor random projection* map according to [Sun+21] is defined as

$$\Pi_{\mathtt{TRP}}(T) := \langle (A_1 \odot \cdots \odot A_n), \mathrm{vec}(T)\rangle \in \mathbb{R}^{d^n \times k},$$

where $\mathrm{vec}(T)$ is the vectorization of T, which takes the coefficients of a tensor in a fixed basis and stacks them into one long vector.

Using this representation reduces the number of random variables from $\mathcal{O}(kd)$ to $\mathcal{O}(k\sqrt[n]{d})$, which is a substantial improvement for large d. Recall that previously introduced Random Fourier Signature Feature kernel, which reduces the computation to a finite-dimensional feature space using the random feature map, is still tensor-valued and thus incurs a polynomial cost in the number of Random Fourier Feature samples n. This is where tensor random projections shine by further reducing the dimensionality in an efficient manner.

Definition 2.12 (Random Fourier Signature Features, Tensor Random Projection) Fix some truncation level $m \in \mathbb{N}$ and take a translation-invariant, continuous, positive-definite kernel $\mathscr{k} : \mathbb{R}^d \times \mathbb{R}^d \to (0;\infty)$ with (α^2, β)-Bernstein spectral measure Λ. Let $W^{(1)}, \dots, W^{(m)} \in \mathbb{R}^{d\times n}$ be Random Fourier Features, with each matrix $W^{(i)} \sim \Lambda$ containing n iid samples for each $i \in [m]$. Further, let $\Pi^{(1)}, \dots, \Pi^{(m)} \in \mathbb{R}^{2n\times n}$ be iid very-sparse random projections with sparsity parameter $2n$. Define the random feature map

$$\widetilde{\varphi}^m_{\mathtt{TRP}}(X) := \sum_{k=0}^{m} \sum_{i\in\Delta_k(|X|)} \langle \Pi^{(1)}, \Delta\widetilde{\varphi}_1 X(t_{i_1})\rangle \odot \cdots \odot \langle \Pi^{(k)}, \Delta\widetilde{\varphi}_k X(t_{i_k})\rangle$$

for all $X \in \mathscr{X}_{\mathbb{R}^d}$ and $\widetilde{\varphi}$ as defined in (2.3). Then the *Very-Sparse Tensor Random Projected Random Fourier Signature Feature* kernel is defined as

$$\widetilde{\mathscr{k}}^m_{\mathtt{TRP}}(X,Y) := \langle \widetilde{\varphi}^m_{\mathtt{TRP}}(X), \widetilde{\varphi}^m_{\mathtt{TRP}}(Y)\rangle_{T^m(\mathbb{R}^d)}$$

for any $X, Y \in \mathscr{X}_{\mathbb{R}^d}$.

Thus, the tensor random projected variant is simply equivalent to applying the Khatri-Rao product of k very-sparse random projections, i.e. the $\Pi_{\mathtt{TRP}}$ operator to the Random Fourier Feature map from (2.3).

To prove a uniform approximation bound, we will need make statements about the tails and moment-boundedness, which requires the introduction of following quasi-norm.

Definition 2.13 (Orlicz Norm) Let X be a real, integrable random variable and $\psi : \mathbb{R} \to \mathbb{R}$ be a non-decreasing, convex function such that $\psi(0) = 0$, then the *Orlicz Norm* of X wrt ψ is given by

$$\|X\|_\psi := \inf\left\{t > 0 : \mathbb{E}\left[\psi\left(\frac{|X|}{t}\right)\right] \leqslant 1\right\}. \tag{2.4}$$

adhering to the convention $\inf \varnothing = \infty$. Throughout the following pages, for any $\alpha > 0$, $\|\cdot\|_{\psi_\alpha}$ denotes the Orlicz Norm wrt to $\psi(x) = \exp(x^\alpha) - 1$ [TOS25].

Strictly speaking, (2.4) is only for $\alpha \geqslant 1$ a well-defined norm, otherwise the triangle inequality does *not* hold. Nonetheless, the above expression is sensible for any $\alpha \in (0;\infty)$, and we choose to call it a norm in these cases as well.

Theorem 2.14 (Pointwise Convergence of Very-Sparse Tensor Random Projected Random Fourier Signature Feature Kernel) *In the setting of Definition 2.12, it holds that for all* $\varepsilon > 0$,

$$\boxed{\mathbb{P}\left(|\widetilde{\mathscr{k}}^m_{\mathtt{TRP}}(X,Y) - \widetilde{\mathscr{k}}^m_{\mathtt{RFF}}(X,Y)| \geqslant \varepsilon\right) \leqslant 6m\Psi(\Lambda)^{dk}\exp(-\min\{\Phi_1(\Lambda,X,Y,\varepsilon), \Phi_2(\Lambda,X,Y,\varepsilon)\})}$$

with constants as defined in (2.20).

Before proving this pointwise bound, we will need the following chain of lemmas.

Corollary 2.15 (Chernoff's Bound) *Let X be a random variable, $\varepsilon > 0$, then*

$$\mathbb{P}(X \geqslant \varepsilon) \leqslant \inf_{\lambda \geqslant 0} \exp\left(-\lambda\varepsilon + \log(\mathbb{E}[\exp(\lambda X)])\right).$$

Proof. See Theorem 2.3.1 in [Ver18]. □

Lemma 2.16 (Young's Inequality) *Let $p, q > 0$ such that $1/p + 1/q = 1$. Then, for every $a, b \geqslant 0$, it holds that*

$$ab \leqslant \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. See Lemma A.3 in [TOS25]. □

Lemma 2.17 (Reverse Young's Inequality) *Let $p, q > 0$ such that $1/p - 1/q = 1$. Then, for every $a \geqslant 0$, $b > 0$, it holds that*

$$ab \geqslant \frac{a^p}{p} - \frac{b^{-q}}{q}.$$

Proof. See Lemma A.4 in [TOS25]. □

Lemma 2.18 *Let X be a real, integrable random variable. Iff $\|X\|_{\psi_\alpha} \leqslant C_\alpha K$ for $\alpha > 0$ and some constant C_α that only depends on α, then*

$$\mathbb{P}(|X| \geqslant \varepsilon|) \leqslant C'_\alpha \exp\left(-\frac{\varepsilon^\alpha}{K^\alpha}\right)$$

for all $\varepsilon > 0$ and some constant C'_α that only depends on C_α.

Proof. Assume wlog $\|X\|_{\psi_\alpha} \leqslant K$, then by definition,

$$\|X\|_{\psi_\alpha} = \inf\left\{t > 0 : \exp\left(\frac{|X|^\alpha}{t^\alpha}\right) \leqslant 2\right\} \leqslant K.$$

It follows by Markov's inequality (2.5) that,

$$\mathbb{P}(|X| \geqslant \varepsilon) \leqslant \exp\left(-\frac{\varepsilon^\alpha}{K^\alpha}\right) \mathbb{E}\left[\exp\left(\frac{\varepsilon^\alpha}{K^\alpha}\right)\right] \leqslant 2\exp\left(-\frac{\varepsilon^\alpha}{K^\alpha}\right)$$

by applying $\varphi(x) = \exp(x^\alpha/K^\alpha)$ on both sides of the inequality. For the converse implication, assume that, there exists $C_{1,\alpha} > 0$ such that

$$\mathbb{P}(|X| \geqslant \varepsilon|) \leqslant C_{1,\alpha} \exp\left(-\frac{\varepsilon^\alpha}{K^\alpha}\right).$$

Then, for some $C_{2,\alpha} > 1$, it follows by the Layer Cake formula that,

$$\mathbb{E}\left[\exp\left(\left(\frac{|X|}{C_{2,\alpha}K}\right)^\alpha\right)\right] = 1 + \int_1^\infty \mathbb{P}\left(\exp\left(\left(\frac{|X|}{C_{2,\alpha}K}\right)^\alpha\right) \geqslant t\right) \mathrm{d}t$$

since $|X|/(C_{2,\alpha}K) \geqslant 0$ a.s. Using a change of variables $u = \exp(t)$, one obtains

$$\begin{aligned}\mathbb{E}\left[\exp\left(\left(\frac{|X|}{C_{2,\alpha}K}\right)^{\alpha}\right)\right] &= 1 + \int_0^{\infty} \exp(u)\mathbb{P}(|X| \geqslant C_{2,\alpha}Ku^{1/\alpha})\,\mathrm{d}u \\ &\leqslant 1 + 2C_{1,\alpha}\int_0^{\infty} \exp\left(u(1 - C_{2,\alpha}^{\alpha})\right)\,\mathrm{d}u \\ &= 1 + \frac{2C_{1,\alpha}}{C_{2,\alpha} - 1}.\end{aligned}$$

The claim follows by choosing $C_{1,\alpha} = 1$ and $C_{2,\alpha} = 3^{1/\alpha}$. □

Theorem 2.19 (Bennet's Inequality) *For a random variable X with $|X - \mathbb{E}[X]| \leqslant C$ a.s. and $\mathrm{Var}(X) = \sigma^2$, it holds that*

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant \varepsilon) \leqslant 2\exp\left(-\frac{\varepsilon^2}{2\sigma^2 + \frac{2}{3}C\varepsilon}\right). \tag{2.5}$$

Proof. Let $\lambda \in \mathbb{R}$. Set $Y := X - \mathbb{E}[X]$. Since $|Y| \leqslant C$ by assumption, it holds that

$$\left|\mathbb{E}\left[Y^k\right]\right| \leqslant \mathbb{E}\left[\left|Y^k\right|\right] \leqslant C^{k-2}\sigma^2.$$

Using the power series expansion and the fact that $\log(1+u) \leqslant u$ for all $u \geqslant 0$, one obtains

$$\log(\mathbb{E}[\exp(\lambda Y)]) \leqslant \sum_{k=2}^{\infty} \frac{\lambda^k}{k!}\left|\mathbb{E}\left[Y^k\right]\right| \leqslant \frac{\sigma^2}{C^2}(\exp(|\lambda|C) - |\lambda|C - 1).$$

Let $\varepsilon > 0$. By applying Chernoff's bound, it follows that

$$\mathbb{P}(Y \geqslant \varepsilon) \leqslant \inf_{\lambda \geqslant 0} \exp\Big(-\lambda\varepsilon + \frac{\sigma^2}{C^2}(\exp(\lambda C) - \lambda C - 1)\Big). \tag{2.6}$$

Using the Taylor series expansion of the exponential, one bounds the second part of the exponent as follows

$$\begin{aligned}\exp(\lambda C) - \lambda C - 1 &\leqslant \frac{(\lambda C)^2}{2} + \sum_{k=3}^{\infty}\frac{(\lambda C)^k}{k!} \leqslant \frac{(\lambda C)^2}{2} + \frac{1}{2}\sum_{k=3}^{\infty}\frac{(\lambda C)^k}{3^{k-2}} \\ &\leqslant \frac{(\lambda C)^2}{2} + \frac{(\lambda C)^2}{2}\sum_{i=1}^{\infty}\frac{(\lambda C)^i}{3^i} = \frac{(\lambda C)^2}{2}\sum_{i=0}^{\infty}\frac{(\lambda C)^i}{3^i}.\end{aligned}$$

For $\lambda C < 3$ the geometric series converges and thus,

$$\exp(\lambda C) - \lambda C - 1 \leqslant \frac{\lambda^2 C^2}{2(1 - \lambda C/3)}. \tag{2.7}$$

Choosing $\lambda \overset{!}{=} \varepsilon/(C(\sigma^2 + \varepsilon/3))$ and plugging into (2.6) yields

$$\mathbb{P}(Y \geqslant \varepsilon) \leqslant \exp\left(-\frac{\varepsilon^2}{2\left(\sigma^2 + C\varepsilon/3\right)}\right). \tag{2.8}$$

One concludes the proof by applying (2.8) to $-Y$ and using a union bound. □

Corollary 2.20 (Bernstein's Inequality) *Let $X_1, \dots, X_n$ iid such that $|X_i| \leqslant C$ a.s. and $\mathrm{Var}(X_i) = \sigma^2/n$ for all $i \in [n]$, then*

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i - \mathbb{E}[X_i]\right| \geqslant \varepsilon\right) \leqslant 2\exp\left(-c\min\left\{\frac{\varepsilon^2}{\sigma^2}, \frac{\varepsilon}{C}\right\}\right)$$

for all $\varepsilon > 0$.

Proof. By Bennet's inequality (2.5),

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i - \mathbb{E}[X_i]\right| \geqslant \varepsilon\right) \leqslant 2\exp\Bigg(-\underbrace{\frac{\varepsilon^2}{2\sigma^2 + \frac{2}{3}C\varepsilon}}_{f(\varepsilon)}\Bigg).$$

Basic arithmetic yields,

$$f(\varepsilon) \gtrsim \frac{\varepsilon^2}{\sigma^2}\mathbb{1}\left\{\varepsilon^2 \leqslant \frac{\sigma^2}{C}\right\} + \frac{\varepsilon}{C}\mathbb{1}\left\{\varepsilon^2 > \frac{\sigma^2}{C}\right\}$$

and subsequently applying the minimum and plugging into (2.20) concludes the proof. □

Lemma 2.21 *Let $X_1, \dots, X_n$ be iid such that for some $\alpha \in (0, 1]$ and all $i \in [n]$, $\|X_i\|_{\psi_\alpha} \leqslant \sigma$, then*

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^n X_i - \mathbb{E}[X_i]\right| \geqslant \varepsilon\right) \leqslant 2\exp\left(-Cn\min\left\{\left(\frac{\varepsilon}{\sigma}\right)^2, \left(\frac{\varepsilon}{\sigma}\right)^\alpha\right\}\right)$$

for all $\varepsilon > 0$ and some constant $C > 0$.

Proof. First, we show that tail and second moment are sufficiently bounded. For any $\varepsilon > 0$ and $i \in [n]$ by Lemma 2.18

$$\mathbb{P}\left(|X_i| \geqslant \varepsilon\right) \leqslant 2\exp\left(-\frac{\varepsilon^\alpha}{\sigma^\alpha}\right). \tag{2.9}$$

For the second moment, it holds that,

$$\mathbb{E}\left[|X_i|^2\right] \leqslant \int_0^\infty \mathbb{P}(|X_i|^2 \geqslant t)\,\mathrm{d}t \leqslant 2\int_0^\infty u\mathbb{P}(|X_i| \geqslant u)\,\mathrm{d}u \leqslant 4\int_0^\infty u\exp\left(-\frac{u^\alpha}{\sigma^\alpha}\right)\mathrm{d}u$$

by using a change of variables and the bound from (2.9). Finally, applying another change of variables yields

$$\mathbb{E}\left[|X_i|^2\right] \leqslant \frac{4\sigma^2}{\alpha}\int_0^\infty t^{\frac{2}{\alpha}-1}\exp(-t)\,\mathrm{d}t = \frac{4\sigma^2}{\alpha}\Gamma\left(\frac{2}{\alpha}\right) = C_\alpha\sigma^2;\ C_\alpha := \frac{4}{\alpha}\Gamma\left(\frac{2}{\alpha}\right). \tag{2.10}$$

Now, fix some threshold $\tau > 0$ and observe that for all $i \in [n]$,

$$X_i = \underbrace{X_i\mathbb{1}\{|X_i| \leqslant \tau\}}_{Y_i} + \underbrace{X_i\mathbb{1}\{|X_i| > \tau\}}_{\xi_i}.$$

Hence, for all $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^n X_i\right| \geqslant \varepsilon\right) \leqslant \sum_{i=1}^n \mathbb{P}\left(|Y_i - \mathbb{E}[Y_i]| \geqslant \frac{\varepsilon}{2}\right) + \sum_{i=1}^n \mathbb{P}\left(|\xi_i - \mathbb{E}[\xi_i]| \geqslant \frac{\varepsilon}{2}\right).$$

By definition, $|Y_i - \mathbb{E}[Y_i]| \leqslant 2\tau$ and furthermore from (2.10) one obtains,

$$\sum_{i=1}^{n} \mathrm{Var}(Y_i) \leqslant \sum_{i=1}^{n} \mathbb{E}\left[X_i^2\right] \leqslant C_\alpha n \sigma^2.$$

Thus, by Bernstein's inequality (2.20), for all $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} Y_i - \mathbb{E}[Y_i]\right| \geqslant \frac{\varepsilon}{2}\right) \leqslant 2\exp\left(-\frac{n\varepsilon^2}{C_\alpha\sigma^2 + \tau\varepsilon}\right). \tag{2.11}$$

By a union bound on the occurrence of a tail and using (2.9) one obtains

$$\mathbb{P}\left(\left|\sum_{i=1}^{n} \xi_i - \mathbb{E}[\xi_i]\right| \geqslant \frac{\varepsilon}{2}\right) \leqslant \mathbb{P}\left(\bigcup_{i=1}^{n} \{|X_i| > \tau\}\right) \leqslant n\mathbb{P}(|X_1| > \tau) \leqslant 2n\exp\left(-\frac{\tau^\alpha}{\sigma^\alpha}\right).$$

Combining both bounds yields

$$\mathbb{P}\left(\left|\sum_{i=1}^{n} X_i - \mathbb{E}[X_i]\right| \geqslant \varepsilon\right) \leqslant 2\exp\left(-\frac{n\varepsilon^2}{C_\alpha\sigma^2 + \tau\varepsilon}\right) + 2n\exp\left(-\frac{\tau^\alpha}{\sigma^\alpha}\right)$$

Setting $\tau \overset{!}{=} C_\alpha \varepsilon n^{1/\alpha} > 0$, one yields for the exponent in (2.11)

$$\eta = \frac{n\varepsilon^2}{C_\alpha\sigma^2 + \tau\varepsilon} = \frac{n\varepsilon^2}{C_\alpha(\sigma^2 + n^{1/\alpha}\varepsilon^2)} \geqslant \frac{1}{2C_\alpha}\left(\frac{n\varepsilon^2}{\sigma^2}\right)$$

by splitting and bounding according to

$$\begin{aligned}\eta &= \eta\mathbb{1}\left\{\eta \leqslant \sigma^2 n^{-1/\alpha}\right\} + \eta\mathbb{1}\left\{\eta > \sigma^2 n^{-1/\alpha}\right\} \\ &\geqslant \frac{1}{2C_\alpha}\left(\frac{n\varepsilon^2}{\sigma^2} + n^{1-1/\alpha}\right) \geqslant \frac{1}{2C_\alpha}\left(\frac{n\varepsilon^2}{\sigma^2}\right).\end{aligned}$$

Thus by combining,

$$\mathbb{P}\left(\left|\sum_{i=1}^{n} X_i - \mathbb{E}[X_i]\right| \geqslant \varepsilon\right) \leqslant 2\exp\left(-C\frac{n\varepsilon^2}{\sigma^2}\right) + 2\exp\left(-C\frac{n\varepsilon^\alpha}{\sigma^\alpha}\right).$$

with $C := \min\{1/2C_\alpha, C_\alpha^\alpha\}$. Using the relation

$$a + b \leqslant 2\exp(-\min\{-\log(a), -\log(b)\}),$$

for $a, b > 0$ concludes the proof. □

Lemma 2.22 *Let $p \sim \Pi$ be a random vector with $p \in \mathbb{R}^n$ sampled from the very-sparse random projection as in Definition 2.10, then for any $v \in \mathbb{R}^n$*

$$\|\langle p, v\rangle\|_{\psi_2} \leqslant C\sqrt{s}\|v\|_2$$

for some constant $C > 0$.

Proof. Observe that, $\mathbb{E}\left[\langle p, v\rangle\right] = 0$, $\mathbb{E}\left[\langle p, v\rangle^2\right] = \|v\|_2^2$ and $p_i v_i \in \{-\sqrt{s}v_i, \sqrt{s}v_i\}$. By Hoeffding's inequality,

$$\mathbb{P}\left(|\langle p, v\rangle| \geqslant \varepsilon\right) \leqslant 2\exp\left(-\frac{\varepsilon^2}{2s\|v\|_2^2}\right)$$

for all $\varepsilon > 0$. Thus, by Lemma 2.18 it follows that,

$$\|\langle p, v\rangle\|_{\psi_2} \leqslant C\sqrt{s}\|v\|_2.$$ □

Lemma 2.23 *Let X, Y be independent random variables such that $\|X\|_{\psi_2}, \|Y\|_{\psi_2} < \infty$, then*

$$\|XY\|_{\psi_1} \leqslant \|X\|_{\psi_2}\|Y\|_{\psi_2}.$$

Proof. By Young's inequality it holds that,

$$\frac{|XY|}{\|X\|_{\psi_2}\|Y\|_{\psi_2}} \leqslant \frac{X^2}{2\|X\|_{\psi_2}^2} + \frac{Y^2}{2\|Y\|_{\psi_2}^2}.$$

Taking the exponential on both sides and subsequently applying the Cauchy-Schwarz-inequality yields

$$\begin{aligned}\mathbb{E}\left[\exp\left(\frac{|XY|}{\|X\|_{\psi_2}\|Y\|_{\psi_2}}\right)\right] &\leqslant \mathbb{E}\left[\exp\left(\frac{X^2}{2\|X\|_{\psi_2}^2} + \frac{Y^2}{2\|Y\|_{\psi_2}^2}\right)\right] \\ &\leqslant \left(\mathbb{E}\left[\exp\left(\frac{X^2}{\|X\|_{\psi_2}^2}\right)\right]\right)^{1/2}\left(\mathbb{E}\left[\exp\left(\frac{Y^2}{\|Y\|_{\psi_2}^2}\right)\right]\right)^{1/2} \leqslant 2,\end{aligned}$$

where the last inequality follows from the Definition 2.13 and likewise the claim. □

Lemma 2.24 *Let $X_1, \ldots, X_n$ be independent random variables such that $\|X_i\|_{\psi_1} < \infty$, then*

$$\left\|\prod_{i=1}^n X_i\right\|_{\psi_{1/n}} \leqslant \prod_{i=1}^n \|X_i\|_{\psi_1}.$$

Proof. By the inequality of arithmetic and geometric means and subsequent exponentiation of both sides, it holds that

$$\exp\left(\left(\prod_{i=1}^n \frac{|X_i|}{\|X_i\|_{\psi_1}}\right)^{1/n}\right) \leqslant \exp\left(\frac{1}{n}\sum_{i=1}^n \frac{|X_i|}{\|X_i\|_{\psi_1}}\right). \tag{2.12}$$

Applying Hölder's inequality to the right hand side yields

$$\mathbb{E}\left[\prod_{i=1}^n \exp\left(\frac{|X_i|}{n\|X_i\|_{\psi_1}}\right)\right] \leqslant \prod_{i=1}^n \left(\mathbb{E}\left[\exp\left(\frac{|X_i|}{\|X_i\|_{\psi_1}}\right)\right]\right)^{1/n} \leqslant 2, \tag{2.13}$$

where the last inequality is due to Definition 2.13. The claim follows by combining the bounds in (2.12) and (2.13) and using Definition 2.13. □

Lemma 2.25 *Let $X_1, \ldots, X_n$ be independent random variables such that for some $\alpha \in (0; 1]$, $\|X_i\|_{\psi_\alpha}$ for all $i \in [n]$, then*

$$\left\|\sum_{i=1}^n X_i\right\|_{\psi_\alpha} \leqslant 2^{(n-1)/\alpha}\sum_{i=1}^n \|X_i\|_{\psi_\alpha}.$$

Proof. See Lemma A.9 in [TOS25]. □

We have now introduced all necessary inequalities to prove the pointwise bound in Theorem 2.14.

Proof of Theorem 2.14. Recall that $m \in \mathbb{N}$ is the truncation level. Then by Equation 3.9 in [TOS25], it holds that

$$\widetilde{\mathscr{k}}^m_{\mathtt{TRP}}(X,Y) = \sum_{k=0}^{m} \zeta_{\mathtt{TRP},\,k};\ \widetilde{\mathscr{k}}^m_{\mathtt{RFF}}(X,Y) = \sum_{k=0}^{m} \zeta_{\mathtt{RFF},\,k}$$

with

$$\zeta_{\mathtt{TRP},k} := \frac{1}{n}\sum_{q=1}^{n} Z_{q,k};\ Z_{q,k} := \sum_{\substack{i\in\triangle_k([|X|])\\ j\in\triangle_k([|Y|])}} \prod_{p=1}^{k} \langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(X(t_{i_p}))\rangle \langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(Y(t_{j_p}))\rangle$$

By linearity of expectation, the fact that the very-sparse tensor random projection is an expected isometry by construction and the independence of its columns it holds that,

$$\begin{aligned}
&\mathbb{E}[Z_{q,k} \,|\, W^{(1)},\dots,W^{(m)}]\\
&= \sum_{\substack{i\in\triangle_k([|X|])\\ j\in\triangle_k([|Y|])}} \prod_{p=1}^{k} \mathbb{E}\left[\langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(X(t_{i_p}))\rangle \langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(Y(t_{j_p}))\rangle \,\middle|\, W_1,\dots,W_m\right]\\
&= \sum_{\substack{i\in\triangle_k([|X|])\\ j\in\triangle_k([|Y|])}} \prod_{p=1}^{k} \mathbb{E}\left[\Pi^{(p)}_{*,q} \odot \Pi^{(p)}_{*,q}\right]^{\intercal} (\Delta\widetilde{\varphi}_p(X(t_{i_p})) \otimes \Delta\widetilde{\varphi}_p(Y(t_{j_p})))\\
&= \sum_{\substack{i\in\triangle_k([|X|])\\ j\in\triangle_k([|Y|])}} \prod_{p=1}^{k} \langle \Delta\widetilde{\varphi}_p(X(t_{i_p})), \Delta\widetilde{\varphi}_p(Y(t_{j_p}))\rangle.
\end{aligned}$$

Thus as in the proof of Theorem C.12 of [TOS25], $\widetilde{\mathscr{k}}^m_{\mathtt{TRP}}$ is conditionally an unbiased estimator of $\widetilde{\mathscr{k}}^m_{\mathtt{RFF}}$ given $W^{(1)},\dots,W^{(m)}$. For any $i \in \triangle_k([|X|])$, $j \in \triangle_k([|Y|])$ by Lemma 2.22 and Lemma 2.23 it holds that,

$$\begin{aligned}
&\|\langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(X(t_{i_p}))\rangle \langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(Y(t_{j_p}))\rangle\|_{\psi_1}\\
&\leqslant \|\langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(X(t_{i_p}))\rangle\|_{\psi_2} \|\langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(Y(t_{j_p}))\rangle\|_{\psi_2}\\
&\leqslant 2C^2 n \|\Delta\widetilde{\varphi}_p(X(t_{i_p}))\|_2 \|\Delta\widetilde{\varphi}_p(Y(t_{j_p}))\|_2.
\end{aligned} \tag{2.14}$$

Independence across the columns within each random projection, (2.14) and Lemma 2.24 give

$$\begin{aligned}
&\left\| \prod_{p=1}^{k} \langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(X(t_{i_p}))\rangle \langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(Y(t_{j_p}))\rangle \right\|_{\psi_{1/m}}\\
&\leqslant C^2 \prod_{p=1}^{k} \|\langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(X(t_{i_p}))\rangle \langle \Pi^{(p)}_{*,q}, \Delta\widetilde{\varphi}_p(Y(t_{j_p}))\rangle\|_{\psi_1}\\
&\leqslant C^2 n^k \prod_{p=1}^{k} \|\Delta\widetilde{\varphi}_p(X(t_{i_p}))\|_2 \|\Delta\widetilde{\varphi}_p(Y(t_{j_p}))\|_2,
\end{aligned} \tag{2.15}$$

where all constants were absorbed into one. Next, by applying Lemma 2.25 and (2.15), we

conclude that,

$$\begin{aligned}\|Z_{q,k}\|_{\psi_{1/k}} &\leqslant C^2 \sum_{\substack{i\in\triangle_k([|X|])\\ j\in\triangle_k([|Y|])}} \left\| \prod_{p=1}^{k} \langle \Pi_{*,q}^{(p)}, \Delta\widetilde{\varphi}_p(X(t_{i_p}))\rangle \langle \Pi_{*,q}^{(p)}, \Delta\widetilde{\varphi}_p(Y(t_{j_p}))\rangle \right\|_{\psi_{1/m}} \\ &\leqslant C^2 n^k 2^{k(|\triangle_k([|X|])|+|\triangle_k([|Y|])|-2)} \sum_{\substack{i\in\triangle_k([|X|])\\ j\in\triangle_k([|Y|])}} \prod_{p=1}^{k} \|\Delta\widetilde{\varphi}_p(X(t_{i_p}))\|_2 \|\Delta\widetilde{\varphi}_p(Y(t_{j_p}))\|_2.\end{aligned} \tag{2.16}$$

Next, by using Corollary C.7 in [TOS25] and (2.16), we observe that,

$$\begin{aligned}\|Z_{q,k}\|_{\psi_{1/k}} &\leqslant C^2 n^k 2^{k(|\triangle_k([|X|])|+|\triangle_k([|Y|])|-2)} \sum_{\substack{i\in\triangle_k([|X|])\\ j\in\triangle_k([|Y|])}} \prod_{p=1}^{k} \|\Delta\widetilde{\varphi}_p(X(t_{i_p}))\|_2 \|\Delta\widetilde{\varphi}_p(Y(t_{j_p}))\|_2 \\ &\leqslant C^2 n^k 2^{k(|\triangle_k([|X|])|+|\triangle_k([|Y|])|-2)} \frac{\|X\|_1^k \|Y\|_1^k}{(k!)^2} \prod_{p=1}^{k} \left(\frac{\|W^{(p)}\|_2}{\sqrt{n}} \right)^2 =: \rho_k(W).\end{aligned} \tag{2.17}$$

Lemma 2.21 and (2.17) yield that, for all $\varepsilon > 0$, $k \in [m] \cup \{0\}$ and some constant $C' > 0$,

$$\mathbb{P}\left(|\zeta_{\mathtt{TRP},\,k} - \zeta_{\mathtt{Sig},\,k}| \geqslant \varepsilon \,\middle|\, W^{(1)}, \ldots, W^{(m)} \right) \leqslant 2\exp\left(-C' n \min\left\{ \frac{\varepsilon^2}{\rho_k(W)^2}, \frac{\varepsilon^{1/k}}{\rho_k(W)^{1/k}} \right\} \right).$$

Define

$$\Xi_k := \frac{\varepsilon (k!)^2}{C^2 n^k 2^{k(|\triangle_k([|Y|])|+|\triangle_k([|Y|])|-2)} \|X\|_1^k \|Y\|_1^k}; \ \Upsilon_k(W) := \prod_{p=1}^{k} \frac{\|W^{(p)}\|_2}{\sqrt{n}}.$$

Assume that $\varepsilon^2/\rho_k(W)^2 \geqslant \varepsilon^{1/k}/\rho_k(W)^{1/k}$.

To obtain an upper bound unconditional of $W^{(1)}, \ldots, W^{(m)}$, we apply the tower rule and the reverse Young's inequality with $p = 1/2$, $q = 1$ to shift the Random Fourier Feature weights into the numerator. Let $\lambda > 0$, which controls the convergence of the geometric series in the moment-generating-function of $W^{(1)}, \ldots, W^{(k)}$, then by Corollary with $p = 1/2$ and $q = 1$

$$\lambda C' n (\Xi_k)^{1/k} \frac{1}{\lambda \Upsilon_k(W)^{2/k}} \geqslant 2(\lambda C' n)^{1/2} (\Xi_k)^{1/2k} - \lambda \Upsilon_k(W)^{2/k}. \tag{2.18}$$

Applying the tower rule to the pointwise conditional upper bound yields

$$\begin{aligned}&\mathbb{P}\left(|\zeta_{\mathtt{TRP},\,k} - \zeta_{\mathtt{RFF},\,k}| \geqslant \varepsilon \,\middle|\, \varepsilon^2/\rho_k(W)^2 \geqslant \varepsilon^{1/k}/\rho_k(W)^{1/k} \right) \\ &= \mathbb{E}\left[\mathbb{P}\left(|\zeta_{\mathtt{TRP},\,k} - \zeta_{\mathtt{RFF},\,k}| \geqslant \varepsilon \,\middle|\, W^{(1)}, \ldots, W^{(m)} \right) \middle| \varepsilon^2/\rho_k(W)^2 \geqslant \varepsilon^{1/k}/\rho_k(W)^{1/k} \right] \\ &\leqslant 2\mathbb{E}\left[\exp\left(-\lambda C' n (\Xi_k)^{1/k} \frac{1}{\lambda \Upsilon_k(W)^{2/k}} \right) \right] \\ &\leqslant 2\mathbb{E}\left[\exp\left(-2(\lambda C' n)^{1/2} (\Xi_k)^{1/2k} + \lambda \Upsilon_k(W)^{2/k} \right) \right] \\ &\leqslant 2\exp\left(-2(\lambda C' n)^{1/2} (\Xi_k)^{1/2k} \right) \mathbb{E}\left[\lambda \Upsilon_k(W)^{2/k} \right],\end{aligned} \tag{2.19}$$

where the next-to-last inequality follows from (2.18). It remains bounding the expectation of the product using the moment-generating function as in Equation C.21 and below from [TOS25]. By choosing $\lambda = 1/(2\beta) < 1/\beta$, we conclude

$$\begin{aligned}\mathbb{E}\left[\lambda \Upsilon_k(W)^{2/k}\right] &\leqslant \mathbb{E}\left[\frac{\lambda}{n}\prod_{p=1}^{k}\|W^{(p)}\|_F^{2/k}\right] \\ &\leqslant \mathbb{E}\left[\frac{\lambda}{nk}\sum_{p=1}^{k}\|W^{(p)}\|_F^{2}\right] \leqslant \left(1+\frac{\alpha^2}{2\beta}+\frac{\alpha^4}{4\beta^2}\right)^d,\end{aligned}$$

where the first inequality is simply using the Frobenius norm definition and the second inequality follows by using the inequality of arithmetic and geometric means, details can be found on page 54 in [TOS25]. This yields

$$\begin{aligned}&\mathbb{P}\left(|\zeta_{\mathtt{TRP},k}-\zeta_{\mathtt{RFF},k}| \geqslant \varepsilon \,\middle|\, \varepsilon^2/\rho_k(W)^2 \geqslant \varepsilon^{1/k}/\rho_k(W)^{1/k}\right) \\ &\leqslant 2\left(1+\frac{\alpha}{2\beta}+\frac{\alpha^2}{4\beta^2}\right)^d \exp\left(-C''\frac{\varepsilon^{1/2k}(k!)^{1/k}}{(\beta\|X\|_1\|Y\|_1)^{1/2}2^{1/2((|\triangle_k([|Y|])|+|\triangle_k([|Y|])|-2))}}\right),\end{aligned}$$

where as usual all constants have been absorbed into C''. Applying the whole procedure to the other case yields

$$\begin{aligned}&\mathbb{P}\left(|\zeta_{\mathtt{TRP},k}-\zeta_{\mathtt{RFF},k}| \geqslant \varepsilon \,\middle|\, \varepsilon^2/\rho_k(W)^2 < \varepsilon^{1/k}/\rho_k(W)^{1/k}\right) \\ &\leqslant 4\left(1+\frac{\alpha}{2\beta}+\frac{\alpha^2}{4\beta^2}\right)^{dk} \exp\left(-C'''\frac{\varepsilon(k!)^2}{\beta^{1/2}n^{k-1/2}2^{k(|\triangle_k([|Y|])|+|\triangle_k([|Y|])|-2)}\|X\|_1^k\|Y\|_1^k}\right)\end{aligned}$$

Set

$$\begin{aligned}\Phi_1(\Lambda,X,Y,\varepsilon) &:= C''\max_{k\in[n]\cup\{0\}}\frac{\varepsilon^{1/2k}(k!)^{1/k}}{(\beta\|X\|_1\|Y\|_1)^{1/2}2^{1/2((|\triangle_k([|Y|])|+|\triangle_k([|Y|])|-2))}} \\ \Phi_2(\Lambda,X,Y,\varepsilon) &:= C'''\max_{k\in[n]\cup\{0\}}\frac{\varepsilon(k!)^2}{\beta^{1/2}n^{k-1/2}2^{k(|\triangle_k([|Y|])|+|\triangle_k([|Y|])|-2)}\|X\|_1^k\|Y\|_1^k} \\ \Psi(\Lambda) &:= 1+\frac{\alpha}{2\beta}+\frac{\alpha^2}{4\beta^2}\end{aligned} \tag{2.20}$$

then using the law of total probability yields the following unconditional bound

$$\mathbb{P}\left(|\zeta_{\mathtt{TRP},k}-\zeta_{\mathtt{RFF},k}| \geqslant \varepsilon\right) \leqslant 6\Psi(\Lambda)^{dk}\exp(-\min\{\Phi_1(\Lambda,X,Y,\varepsilon),\Phi_2(\Lambda,X,Y,\varepsilon)\}).$$

Applying the union bound concludes the proof,

$$\mathbb{P}\left(|\widetilde{\mathscr{k}}^m_{\mathtt{TRP}}(X,Y)-\widetilde{\mathscr{k}}^m_{\mathtt{RFF}}(X,Y)| \geqslant \varepsilon\right) \leqslant 6m\Psi(\Lambda)^{dk}\exp(-\min\{\Phi_1(\Lambda,X,Y,\varepsilon),\Phi_2(\Lambda,X,Y,\varepsilon)\}).$$

□

Note that we gave an upper bound on the approximation error between the Random Fourier Signature Feature kernel and the Very-Sparse Tensor Random Projection variant. To get a bound for the discretized signature kernel we use a union bound with Theorem 2.9. This yields the following

Corollary 2.26 *In the setting of Definition 2.12, it holds that for all* $\varepsilon > 0$,

$$\begin{aligned}
&\mathbb{P}(|\widetilde{\mathscr{k}}_{\mathtt{TRP}}(X,Y) - \widetilde{\mathscr{k}}_{\mathtt{Sig}}(X,Y)| \geqslant \varepsilon) \\
&\leqslant 6m\Psi(\Lambda)^{dk}\exp(-\min\{\Phi_1(\Lambda,k,\varepsilon/2),\Phi_2(\Lambda,k,\varepsilon/2)\}) + \\
&m\theta_{n,d,I}\left(\left(\frac{2\tau_{d,m,V}}{\varepsilon}\right)^{d/(d+1)} + d + 1\right)\exp\left(-\xi_{n,d,\Lambda}\min\left\{\left(\frac{\varepsilon}{2\tau_{d,m,V}}\right)^2, \left(\frac{\varepsilon}{2\tau_{d,m,V}}\right)^{1/m}\right\}\right),
\end{aligned}$$

where the constants have been defined in the respective theorems.

This corollary also concludes the thesis. We use this closing paragraph to comment on the improved runtime compared to the iid gaussian random projections that were chosen in Theorem 3.7 of [TOS25]. Recall that we set the sparsity parameter to $2n$. The very-sparse random projection reduces the computation of the projected increments in Algorithm D.3 of [TOS25] from $\mathscr{O}(dn)$ to $\mathscr{O}(dn/\sqrt{2n})$. Thus, the total runtime reduces from $\mathscr{O}(mNL_{\max}n(d+n))$ as stated in Remark 3.9 [TOS25] to $\mathscr{O}(mNL_{\max}(d\sqrt{n}+n^2))$. Asymptotically that is equivalent to

$$\Sigma = \frac{dn+n^2}{d\sqrt{n}+n^2} = \sqrt{n}\left(\frac{1+n/d}{1+n^{3/2}/d}\right),$$

which means given $n \ll d$,

$$\Sigma = \sqrt{n}\left(\frac{1+\mathscr{o}(1)}{1+\mathscr{o}(1)}\right) = \sqrt{n}$$

or a speed-up of order $\sqrt{n}$. Thus, we expect fair speed-up rates for time series with high dimensionality, which do occur in natural language processing with one-hot-encoded tokens or genome data.

We also want to note that also a uniform bound can be proven using generic chaining [Tal05], [Dir15] and similar tail bounds as in the pointwise case. The only requirement is that the 1-variation is bounded by some constant $V > 0$ to build a deterministic upper bound on the diameter and the $\gamma_{1/k}$-functional $\gamma_{1/k}$[Tal05], which is estimated through Dudley's Entropy Integral [Dir15]. However, we choose to omit the proof due to time constraints.

List of Abbreviations and Symbols

$\gamma_{1/k}$ Talagrand's $\gamma_{1/k}$-functional. 31

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